Name

Mathematics Stage 5 Path

Further Expansion and Factorisation

Book 1 Expand, factorise and simplify algebraic expressions including special products

Version: 261007

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Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

MA5-ALG-P-02 selects and applies appropriate algebraic techniques to operate with algebraic fractions, and expands, factorises and simplifies algebraic expressions

Expand, factorise and simplify algebraic expressions including special products

- Prove and apply these special products $(a-b)(a+b)=a^2-b^2$, $(a+b)^2=a^2+2ab+b^2$, $(a-b)^2=a^2-2ab+b^2$
- Expand and simplify a variety of algebraic expressions including binomial products and the special products
- Factorise algebraic expressions involving the special products and strategies, including common factors, grouping in pairs for 4-term expressions and quadratic trinomials (monic and non-monic)

Expanding Perfect Squares

Review

☐ Expand

a $(x+3)(x+3)$	**b** $(x+5)(x+5)$
c $(x-6)(x-6)$	**d** $(x-4)^2$
e $(3x+5)^2$	**f** $(5x-7)^2$

☐ Simplify

a $(3)^2$	**b** $(5)^2$	**c** $(-4)^2$	**d** $(-7)^2$
e $(3x)^2$	**f** $(-5y)^2$	**g** $(7x)^2$	**h** $(6xy)^2$

☐ Find $2ab$ given a and b

a $a: x \quad b: 3$	**b** $a: y \quad b: 4$	**c** $a: x \quad b: -3$	**d** $a: 3x \quad b: -5$
$2ab =$	$2ab =$	$2ab =$	$2ab =$

ⓘ Perfect Squares

A perfect square is an expression of the form $(a+b)^2$.

We identify the two terms inside the brackets as a and b.

ⓘ Identify a and b for a perfect square

$(x+3)^2$	$(2x+3)^2$	$(5x-3y)^2$
a: x $\quad$ b: 3	a: $2x$ $\quad$ b: 3	a: $5x$ $\quad$ b: $-3y$

Identify a and b for these perfect squares

a $(x+7)^2$	**b** $(5x+7)^2$	**c** $(5x+7y)^2$
a: $\quad$ b:	a: $\quad$ b:	a: $\quad$ b:
d $(x-3)^2$	**e** $(4x-3y)^2$	**f** $(1-6x)^2$
a: $\quad$ b:	a: $\quad$ b:	a: $\quad$ b:

ⓘ Identify a and b for a perfect square

Compute $2ab$ for these perfect squares

$(x+3)^2$	$(2x+3)^2$	$(2x-3y)^2$
$2ab = 2(x)(3)$ $= 6x$	$2ab = 2(2x)(3)$ $= 12x$	$2ab = 2(2x)(-3y)$ $= -12xy$
a $(x+7)^2$	**b** $(5x+7)^2$	**c** $(5x+7y)^2$
$2ab =$	$2ab =$	$2ab =$
d $(x+7)^2$	**e** $(5x+7)^2$	**f** $(5x+7y)^2$
$2ab =$	$2ab =$	$2ab =$

Perfect Square Expansion

$$(a+b)^2 = a^2 + 2ab + b^2$$

1. Identify a and b.
2. Write the expression $a^2 + 2ab + b^2$ using a and b from step 1.
3. Simplify.

Example Expand perfect squares

$(x+4)^2$

1. $a = x \quad b = 4$
2. $(x)^2 + 2(x)(4) + (4)^2$
3. $x^2 + 8x + 16$

$(5x-3y)^2$

1. $a = 5x \quad b = -3y$
2. $(5x)^2 + 2(5x)(-3y) + (-3y)^2$
3. $25x^2 - 30xy + 9y^2$

Guided Practice

Expand these expressions

a $(x+y)^2$

$x^2 + 2xy + y^2$

b $(x+3)^2$

$x^2 + 6x + 9$

c $(x-7)^2$

$x^2 - 14x + 49$

d $(3x+1)^2$

$9x^2 + 6x + 1$

e $(6x-5)^2$

$36x^2 - 60x + 25$

f $(8y+3x)^2$

$64y^2 + 48xy + 9x^2$

Key Ideas

1. Expressions of the form $(a+b)^2$ are called …… ……
2. To quickly expand perfect squares, follow the rule: $(a+b)^2 =$ ……

1 Expand

a $(x+1)^2$

$x^2 + 2x + 1$

b $(x+2)^2$

$x^2 + 4x + 4$

c $(x+3)^2$

$x^2 + 6x + 9$

d $(x+4)^2$

$x^2 + 8x + 16$

e $(x+10)^2$

$x^2 + 20x + 100$

f $(x+n)^2$

$x^2 + 2nx + n^2$

2 Expand these perfect squares.

a $(x+3)^2$

$x^2 + 6x + 9$

b $(x-3)^2$

$x^2 - 6x + 9$

c $(x+7)^2$

$x^2 + 14x + 49$

d $(x-2)^2$

$x^2 - 4x + 4$

e $(x-6)^2$

$x^2 - 12x + 36$

f $(x-7)^2$

$x^2 - 14x + 49$

g $(x-4)^2$

$x^2 - 8x + 16$

h $(x-12)^2$

$x^2 - 24x + 144$

i $(x-9)^2$

$x^2 - 18x + 81$

3 Jimmy expanded $(x+5)^2 = x^2 + 25$. Explain why he is incorrect.

He is missing the middle term $x^2 + 10x + 25$

4 Fill in the missing numbers.

a $(x+3)^2 = x^2 + \square x + \square$

$6, 9$

b $(x-7)^2 = x^2 - 14x + \square$

49

c $(x-2)^2 = x^2 __ \square + \square$

$-4x, 4$

d $(x - \square)^2 = x^2 - 6x + \square$

$3, 9$

e $(x - \square)^2 = x^2 - 10x + \square$

$5, 25$

f $(x + \square)^2 = x^2 + 12x + \square$

$6, 36$

g $(x + \square)^2 = x^2 __ \square + 49$

$7, +14x$

h $(x + \square)^2 = x^2 + 18x + \square$

$9, 81$

5 Expand these perfect squares.

a $(3-x)^2$

$9 - 6x + x^2$

b $(5-x)^2$

$25 - 10x + x^2$

c $(1-x)^2$

$1 - 2x + x^2$

d $(11-x)^2$

$121 - 22x + x^2$

e $(7-x)^2$

$49 - 14x + x^2$

f $(8-x)^2$

$64 - 16x + x^2$

6 Expand

a $(x+5)^2$

$x^2 + 10x + 25$

b $(x-5)^2$

$x^2 - 10x + 25$

c $(-x+5)^2$

$x^2 - 10x + 25$

d $(-x-5)^2$

$x^2 + 10x + 25$

7 Explain why $(3-x)^2$ and $(x-3)^2$ is the same.

$(3-x)^2 = 9 - 6x + x^2$ and $(x-3)^2 = x^2 - 6x + 9$, which are the same.

This is because you can factorise a negative out of the inner terms, which disappears when squared.

$(3-x)^2 = (-(x-3))^2 = (x-3)^2$

8 Expand

a $(x+3)^2$

$x^2 + 6x + 9$

b $(2x+3)^2$

$4x^2 + 12x + 9$

c $(3x+3)^2$

$9x^2 + 18x + 9$

d $(4x+3)^2$

$16x^2 + 24x + 9$

e $(10x+3)^2$

$100x^2 + 60x + 9$

9 Expand these perfect squares.

a $(2x+3)^2$

$4x^2 + 12x + 9$

b $(2x+5)^2$

$4x^2 + 20x + 25$

c $(3x+2)^2$

$9x^2 + 12x + 4$

d $(5x+2)^2$

$25x^2 + 20x + 4$

e $(7+2x)^2$

$49 + 28x + 4x^2$

f $(5+3x)^2$

$25 + 30x + 9x^2$

g $(5x-2)^2$

$25x^2 - 20x + 4$

h $(7-2x)^2$

$49 - 28x^2 + 4x^2$

i $(5-3x)^2$

$25 - 30x + 9x^2$

j $(2-3x)^2$

$4 - 12x + 9x^2$

k $(9-2x)^2$

$81 - 36x + 4x^2$

l $(10-4x)^2$

$100 - 80x + 16x^2$

m $(3x+5y)^2$

$9x^2 + 30xy + 25y^2$

n $(7x+3y)^2$

$49x^2 + 42xy + 9y^2$

o $(4x-9y)^2$

$16x^2 - 72xy + 81y^2$

p $(3x-10y)^2$

$9x^2 - 60xy + 100y^2$

q $(4x-6y)^2$

$16x^2 - 48xy + 36y^2$

r $(6x+5y)^2$

$36x^2 + 60xy + 25y^2$

10 Expand

a $(x+y)^2$

$x^2 + 2xy + y^2$

b $(x+2y)^2$

$x^2 + 4xy + 4y^2$

c $(x+3y)^2$

$x^2 + 6xy + 9y^2$

d $(2x+3y)^2$

$4x^2 + 12xy + 9y^2$

e $(3x+5y)^2$

$9x^2 + 30xy + 25y^2$

11 Expand

a $(3x + 5y)^2$

$9x^2 + 30xy + 25y^2$

b $(3x - 5y)^2$

$9x^2 - 30xy + 25y^2$

c $(5x + 3y)^2$

$25x^2 + 30xy + 9y^2$

d $(5x - 3y)^2$

$25x^2 - 30xy + 9y^2$

12 Expand the following.

a $2(x + 3)^2$

$2x^2 + 12x + 18$

b $4(m + 5)^2$

$4m^2 + 40m + 100$

c $2(a - 7)^2$

$2a^2 - 28a + 98$

d $-3(y + 5)(y + 5)$

$-3y^2 - 30y - 75$

e $3(2b - 1)^2$

$12b^2 - 12b + 3$

f $-3(2y - 6)^2$

$-12y^2 + 72y - 108$

13 Expand the following perfect squares.

a $\left(x + \frac{1}{3}\right)^2$

$x^2 + \frac{2x}{3} + \frac{1}{9}$

b $\left(x - \frac{2}{5}\right)^2$

$x^2 - \frac{4x}{5} + \frac{4}{25}$

c $\left(3x + \frac{5}{7}\right)^2$

$9x^2 + \frac{30x}{7} + \frac{25}{49}$

d $\left(3x + \frac{2}{5}\right)^2$

$9x^2 + \frac{12x}{5} + \frac{4}{25}$

e $\left(\frac{x}{5} - \frac{3}{8}\right)^2$

$\frac{x^2}{25} - \frac{3x}{20} + \frac{9}{64}$

f $\left(\frac{2x}{3} + \frac{9y}{8}\right)^2$

$\frac{4x^2}{9} + \frac{3xy}{2} + \frac{81y^2}{64}$

14 Expand

a $(x^2 + 3)^2$

$x^4 + 6x^2 + 9$

b $(x^2 - 3)^2$

$x^4 - 6x^2 + 9$

c $(x^3 + 2)^2$

$x^6 + 4x^3 + 4$

d $(2x^2 + 5)^2$

$4x^4 + 20x^2 + 25$

e $(x^2 + y^2)^2$

$x^4 + 2x^2y^2 + y^4$

15 Expand and simplify

a $2(x+3)^2 - 18$

$2x^2 + 12x$

b $(x+4)^2 - (x-4)^2$

$16x$

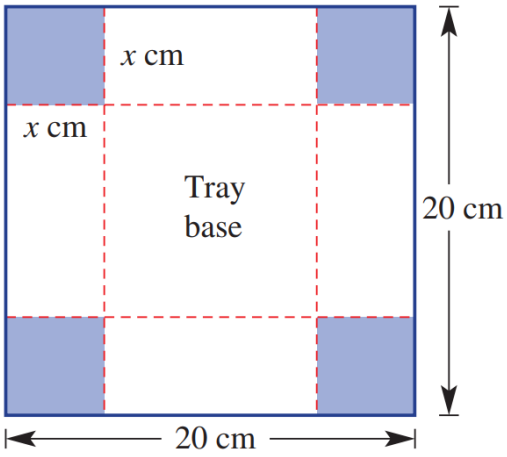
c $(x+a)^2 - (x-a)^2$

$4ax$

d $(x+1)^2 + (x-1)^2$

$2x^2 + 2$

16 A square piece of tin of side length 20 cm has four squares of side length x cm removed from each corner. The sides are folded up to form a tray. The centre square forms the tray base.



a Write an expression for the side length of the base of the tray.

$20 - 2x$

b Write an expression for the area of the base of the tray. Expand your answer.

$(20-2x)^2 = 400 - 80x + 4x^2$

c Find the area of the tray base if $x = 3$.

196 cm^2

Expanding Difference of Two Squares

Review

☐ Expand

a $(x+3)(x-3)$	**b** $(x+5)(x-5)$
c $(x-6)(x+6)$	**d** $(3x-4)(3x+4)$
e $(y-11)(y+11)$	**f** $(2x-5)(5+2x)$

Difference of Two Squares

$$(a-b)(a+b) = a^2 - b^2$$

1. Identify a and b.
2. Write the expression $a^2 - b^2$ using a and b from step 1.
3. Simplify.

Example Expand difference of two squares

$(y-3)(y+3)$	$(6-2t)(6+2t)$
1. $a = y \quad b = 3$	1. $a = 6 \quad b = 2t$
2. $(y)^2 - (3)^2$	2. $(6)^2 - (2t)^2$
3. $y^2 - 9$	3. $36 - 4t^2$

Guided Practice

Expand these expressions

a $(x-4)(x+4)$ $x^2 - 16$	**b** $(x+8)(x-8)$ $x^2 - 64$
c $(2x+3)(2x-3)$ $4x^2 - 9$	**d** $(7-3x)(7+3x)$ $49 - 9x^2$
e $(x^3+5)(x^3-5)$ $x^6 - 25$	**f** $(3x^2-5)(3x^2+5)$ $9x^4 - 25$

Key Ideas

1. Expressions in the form $(a+b)(a-b)$ are called ....
2. To quickly expand, follow the rule: $(a+b)(a-b) =$ .....

1 Expand these differences of two squares.

a $(x+1)(x-1)$ x^2-1	**b** $(x+3)(x-3)$ x^2-9	**c** $(x+8)(x-8)$ x^2-64
d $(x-6)(x+6)$ x^2-36	**e** $(6-x)(6+x)$ $36-x^2$	**f** $(5+x)(5-x)$ $25-x^2$
g $(x+11)(x-11)$ x^2-121	**h** $(12+x)(12-x)$ $(144-x^2)$	**i** $(7+x)(7-x)$ $49-x^2$

2 Expand

a $(x+3)(x-3)$ x^2-9	**b** $(2x+3)(2x-3)$ $4x^2-9$	**c** $(3x+3)(3x-3)$ $9x^2-9$	**d** $(5x+3)(5x-3)$ $25x^2-9$

3 Expand

a $(x+4)(x-4)$ x^2-16	**b** $(x-4)(x+4)$ x^2-16	**c** $(4+x)(4-x)$ $16-x^2$	**d** $(4-x)(4+x)$ $16-x^2$

4 Expand these differences of two squares.

a $(3x-2)(3x+2)$ $9x^2-4$	**b** $(5x-4)(5x+4)$ $25x^2-16$	**c** $(4x-3)(4x+3)$ $16x^2-9$
d $(9x-5y)(9x+5y)$ $81x^2-25y^2$	**e** $(11x-y)(11x+y)$ $121x^2-y^2$	**f** $(8x+2y)(8x-2y)$ $64x^2-4y^2$
g $(7x-5y)(7x+5y)$ $49x^2-25y^2$	**h** $(8x-3y)(8x+3y)$ $64x^2-9y^2$	**i** $(9x-4y)(9x+4y)$ $81x^2-16y^2$
j $(x^2+7)(x^2-7)$ x^4-49	**k** $(x^2-11)(x^2+11)$ x^4-121	**l** $(x^3+2y)(x^3-2y)$ x^6-4y^2
m $(2x^3+3y^2)(2x^3-3y^2)$ $4x^6-9y^4$	**n** $(5x^2-14x)(5x^2+14x)$ $25x^4-196x^2$	**o** $(3x^8+5y^3)(3x^8-5y^3)$ $9x^{16}-25y^6$

5 Expand

a $(x+y)(x-y)$ x^2-y^2

b $(2x+y)(2x-y)$ $4x^2-y^2$

c $(x+3y)(x-3y)$ x^2-9y^2

d $(2x+3y)(2x-3y)$ $4x^2-9y^2$

e $(5x+7y)(5x-7y)$ $25x^2-49y^2$

6 Expand

a $(x+3)(x-3)$ x^2-9

b $(x+3)(x+3)$ x^2+6x+9

c $(x-3)(x-3)$ x^2-6x+9

d $(x-3)(x+3)$ x^2-9

e $(3-x)(3+x)$ $9-x^2$

f $(3+x)(3-x)$ $9-x^2$

7 Expand

a $(x+3)(x-3)$ x^2-9

b $(x^2+3)(x^2-3)$ x^4-9

c $(x^3+3)(x^3-3)$ x^6-9

d $(x^4+3)(x^4-3)$ x^8-9

e $(x^n+3)(x^n-3)$ $x^{2n}-9$

8 Expand

a $(x^2+5)(x^2-5)$ x^4-25

b $(2x^2+5)(2x^2-5)$ $4x^4-25$

c $(x^3+2y)(x^3-2y)$ x^6-4y^2

d $(2x^3+3y^2)(2x^3-3y^2)$ $4x^6-9y^4$

e $(3x^4+4y^3)(3x^4-4y^3)$ $9x^8-16y^6$

9 Expand these differences of two squares involving fractional terms.

a $\left(x+\frac{1}{3}\right)\left(x-\frac{1}{3}\right)$ $x^2-\frac{1}{9}$

b $\left(x+\frac{3}{4}\right)\left(x-\frac{3}{4}\right)$ $x^2-\frac{9}{16}$

c $\left(\frac{x}{3}+\frac{4}{5}\right)\left(\frac{x}{3}-\frac{4}{5}\right)$ $\frac{x^2}{9}-\frac{16}{25}$

d $\left(\frac{x}{2}+\frac{y}{7}\right)\left(\frac{x}{2}-\frac{y}{7}\right)$ $\frac{x^2}{4}-\frac{y^2}{49}$

e $\left(2x-\frac{3}{7}\right)\left(2x+\frac{3}{7}\right)$ $4x^2-\frac{9}{49}$

f $\left(\frac{7}{3}-5x\right)\left(\frac{7}{3}+5x\right)$ $\frac{49}{9}-25x^2$

10 Expand these differences of two squares.

a $(x+\sqrt{7})(x-\sqrt{7})$

x^2-7

b $(x-\sqrt{5})(x+\sqrt{5})$

x^2-5

c $(3x+\sqrt{17})(3x-\sqrt{17})$

$9x^2-17$

d $(x^2+\sqrt{21})(x^2-\sqrt{21})$

x^4-21

e $(2x^3-\sqrt{10})(2x^3+\sqrt{10})$

$4x^6-10$

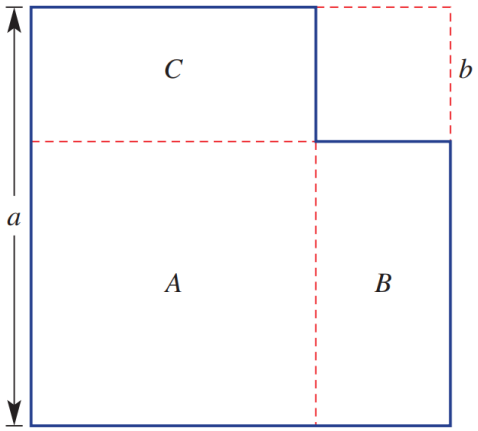
f $(\sqrt{6}+3x)(\sqrt{6}-3x)$

$6-9x^2$

11 A square of side length x has one side reduced by 1 unit and the other increased by 1 unit. By forming an expression for the area, explain why the area of the resulting rectangle is not the same as the original square.

$(x+1)(x-1)=x^2-1$. The new area is 1 less than the original.

12 A square of side length b is removed from a square of side length a.



a Using subtraction, write down an expression for the remaining area.

a^2-b^2

b Write expressions for the area of the regions:

i A

$(a-b)^2$

ii B

$b(a-b)$

iii C

$b(a-b)$

c Add all the expressions from part **b** and confirm you get your answer from part **a**.

13 Compute the following using the difference of two squares.

a $27\times 33=(30-3)\times(30+3)$

$=30^2-\square^2$

$=900-_____$

$=$

891

b $78\times 82=(80-_____)\times(80+_____)$

6396

c 96×104

9984

d 189×211

39879

14 Compute the following using perfect squares.

a $53^2 = (50+3)^2$

$= 50^2 + 2 \times \square \times \square + 3^2$

$=$

2809

b $98^2 = (100-2)^2$

$= \square - 2 \times 100 \times \square + 2^2$

$=$

9604

c 82^2

6724

d 39^2

1521

e 102^2

10404

f 197^2

38809

15 Which is larger? 77×77 or 78×76?
Compute each using perfect squares and the difference of two squares.

a 77×77

5929

b 78×76

5928

16 Factorise using the difference of two squares.

a $x^2 - \frac{1}{9}$

$\left(x + \frac{1}{3}\right)\left(x - \frac{1}{3}\right)$

b $x^2 - \frac{3}{4}$

$\left(x + \frac{\sqrt{3}}{2}\right)\left(x - \frac{\sqrt{3}}{2}\right)$

c $x^2 - \frac{7}{16}$

$\left(x + \frac{\sqrt{7}}{4}\right)\left(x - \frac{\sqrt{7}}{4}\right)$

d $x^2 - \frac{5}{16}$

$\left(x + \frac{\sqrt{5}}{4}\right)\left(x - \frac{\sqrt{5}}{4}\right)$

e $(x-2)^2 - 20$

$(x - 2 + 2\sqrt{5})(x - 2 - 2\sqrt{5})$

f $(x+4)^2 - 27$

$(x + 4 + 3\sqrt{3})(x + 4 - 3\sqrt{3})$

g $(x-7)^2 - 40$

$(x - 7 + 2\sqrt{10})(x - 7 - 2\sqrt{10})$

h $3x^2 - 4$

$(\sqrt{3}x + 2)(\sqrt{3}x - 2)$

i $5x^2 - 9$

$(\sqrt{5}x + 3)(\sqrt{5}x - 3)$

j	$5x^2 - 120$	$5(x + 2\sqrt{6})(x - 2\sqrt{6})$	**k**	$2(x + 3)^2 - 10$	$2(x + 3 + \sqrt{5})(x + 3 - \sqrt{5})$	**l**	$2x^2 - 96$	$2(x + 4\sqrt{3})(x - 4\sqrt{3})$
m	$3(x - 1)^2 - 21$	$3(x - 1 + \sqrt{7})(x - 1 - \sqrt{7})$	**n**	$4(x - 4)^2 - 48$	$4(x - 4 + 2\sqrt{3})(x - 4 - 2\sqrt{3})$	**o**	$5(x + 6)^2 - 90$	$5(x + 6 + 3\sqrt{2})(x + 6 - 3\sqrt{2})$
p	$9 - (x + 3)^2$	$-x(x + 6)$	**q**	$16 - (x + 4)^2$	$-x(x + 8)$	**r**	$25 - (x - 5)^2$	$x(10 - x)$
s	$49 - (x - 1)^2$	$(x + 6)(8 - x)$	**t**	$(x + 2)^2 - (x + 3)^2$	$-(2x + 5)$	**u**	$(y - 7)^2 - (y + 4)^2$	$-11(2y - 3)$
v	$(b + 5)^2 - (b - 5)^2$	$20b$	**w**	$(a + 3)^2 - (a - 5)^2$	$16(a - 1)$	**x**	$(2w + 3x)^2 - (3w + 4x)^2$	$-(5w + 7x)(w + x)$

Mixed Expansion Practice

ⓘ Expansion

Expanding Single Brackets	Binomial Expansion
$a(b+c) = ab + ac$	$(a+b)(c+d) = ac + ad + bc + bd$
Perfect Square	**Difference of Two Squares**
$(a+b)^2 = a^2 + 2ab + b^2$	$(a-b)(a+b) = a^2 - b^2$

FOUNDATION

1 Match the equivalent expressions.

a	$x(x+4)$	**A**	$x^2 - 8x + 16$
b	$(x+1)(x+4)$	**B**	$x^2 + 5x + 4$
c	$(x+4)^2$	**C**	$x^2 + 4x$
d	$(x-4)^2$	**D**	$x^2 - 16$
e	$(x+4)(x-4)$	**E**	$x^2 + 8x + 16$

a - C, b - B, c - E, d - A, e - D

2 Expand

a	$(x+3)(x+4)$ $x^2 + 7x + 12$	**b**	$(x+3)(x-4)$ $x^2 - x - 12$	**c**	$(x-3)(x+4)$ $x^2 + x - 12$
d	$(x-3)(x-4)$ $x^2 - 7x + 12$	**e**	$(x+3)(x+3)$ $x^2 + 6x + 9$	**f**	$(x+3)(x-3)$ $x^2 - 9$
c	$x(x+5)$ $x^2 + 5x$	**d**	$(x+5)(x-5)$ $x^2 - 25$	**e**	$(x+5)^2$ $x^2 + 10x + 25$
f	$(x-5)^2$ $x^2 - 10x + 25$	**g**	$(x-5)(x+5)$ $x^2 - 25$	**h**	$(x-5)(y+5)$ $xy + 5x - 5y - 25$
i	$(x-10)^2$ $x^2 - 20x + 100$	**j**	$(x+7)^2$ $x^2 + 14x + 49$	**k**	$(x+8)(x-8)$ $x^2 - 64$
l	$(x+11)(x-11)$ $x^2 - 121$	**m**	$(x+11)(x+11)$ $x^2 + 22x + 121$	**n**	$(x-7)(x-7)$ $x^2 - 14x + 49$
o	$(x+12)(x-12)$ $x^2 - 144$	**p**	$(x-3)(x-3)$ $x^2 - 6x + 9$	**q**	$(x+6)(x-6)$ $x^2 - 36$
r	$(x+12)^2$ $x^2 + 24x + 144$	**s**	$(x-12)^2$ $x^2 - 24x + 144$	**t**	$(x-6)^2$ $x^2 - 12x + 36$

3 Expand

a $(2x + 3)(2x + 5)$

$4x^2 + 16x + 15$

b $(2x + 3)(2x - 5)$

$4x^2 - 4x - 15$

c $(2x + 3)(2x + 3)$

$4x^2 + 12x + 9$

d $(2x + 3)(2x - 3)$

$4x^2 - 9$

e $(2x - 3)(2x - 3)$

$4x^2 - 12x + 9$

4 Expand the following.

a $2(x + 3)^2$

$2x^2 + 12x + 18$

b $4(m + 5)^2$

$4m^2 + 40m + 100$

c $2(a - 7)^2$

$2a^2 - 28a + 98$

d $-3(y + 5)(y + 5)$

$-3y^2 - 30y - 75$

e $3(2b - 1)^2$

$12b^2 - 12b + 3$

f $-3(2y - 6)^2$

$-12y^2 + 72y - 108$

g $2(x - 3)^2$

$2x^2 - 12x + 18$

h $-2(x + 8)(x - 8)$

$-2x^2 + 128$

i $-4(x + 9)(x + 9)$

$-4x^2 - 72x - 324$

j $3(x + 5)(x - 3)$

$3x^2 + 6x - 45$

k $-3(a + 2)(a - 7)$

$-3a^2 + 15a + 42$

l $-2(x - 7)(x + 7)$

$-2x^2 + 98$

m $3(2x + 3)(2x + 3)$

$12x^2 + 36x + 27$

n $6(3x - 4)(3x + 4)$

$54x^2 - 96$

o $-2(x + 4)(3x - 7)$

$-6x^2 - 10x + 56$

5 Expand and simplify the following.

a $3(x+2)$

$3x+6$

b $x(x-1)$

x^2-x

c $-2(3-x)$

$-6+2x$

d $-2x(5-3x)$

$-10x+6x^2$

e $x^2(x-3)$

x^3-3x^2

f $x^2(x^2-3x)$

x^4-3x^2

g $3xy(3x-y)$

$9x^2y-3xy^2$

h $-3xy(4y^2-2xy)$

$-12xy^3+6x^2y^2$

i $(x+1)(x+2)$

x^2+3x+2

j $(x-3)(x+2)$

x^2-x-6

k $(x-5)(x-4)$

$x^2-9x+20$

l $2(x+1)(x-2)$

$2x^2-2x-4$

m $x(x-3)(x+2)$

x^3-x^2-6x

n $2x(3-x)(x^2+2)$

$6x^3+12x-2x^4-4x^2$

o $3x^2(3x^2-5)(4-x^2)$

$51x^4-9x^6-60x^2$

p $-(x+3)(2-x)$

x^2+x-6

q $(x+3)(x^2+3)$

$x^3+3x+3x^2+9$

r $(2x-3)(x^2-3)$

$2x^3-6x-3x^2+9$

s $(x+1)^2$

x^2+2x+1

t $(2-x)^2$

$4-4x+x^2$

u $-(2x^2-3)^2$

$-4x^4+12x^2-9$

v $-2x(x^2+2)^2$

$-2x^5-8x^3-8x$

w $\left(x+\frac{1}{x}\right)^2$

$x^2+2+\frac{1}{x^2}$

x $\left(x+\frac{1}{\sqrt{x}}\right)^2$

$x^2+2\sqrt{x}+\frac{1}{x}$

6 All of these are common mistakes students make when expanding.
Identify the mistake in each set of working, then give the correct answer.

a $-x(x-7) = -x^2 - 7x$



$-x^2 + 7x$

b $3a - 7(4-a) = 3a - 28 - 7a$

$= -4a - 28$



$10a - 28$

c $(2x+3)^2 = 4x^2 + 9$



$4x^2 + 12x + 9$

d $(x+2)^2 - (x^2+4) = 0$



$4x$

7 Expand and simplify the following.

a $\left(t + \frac{1}{t}\right)^2$

$t^2 + 2 + \frac{1}{t^2}$

b $\left(t - \frac{1}{t}\right)^2$

$t^2 - 2 + \frac{1}{t^2}$

c $\left(t + \frac{1}{t}\right)\left(t - \frac{1}{t}\right)$

$t^2 - \frac{1}{t^2}$

d $\left(\frac{4}{m} - m\right)\left(\frac{4}{m} + m\right)$

$\frac{16}{m^2} - m^2$

e $\left(xy - \frac{1}{y}\right)\left(xy + \frac{1}{y}\right)$

$x^2y^2 - \frac{1}{y^2}$

f $\left(\frac{4a}{3} + 2\right)\left(\frac{4a}{3} - 2\right)$

$\frac{16a^2}{9} - 4$

8 Expand and simplify the following.

a $3a^2(4a^3 - b^4) + 2a^2(5a^3 + 3b^4)$

$22a^5 + 3a^2b^4$

b $2y^3(5y^2 + 3y^4) - 4y^2(2y^2 - y^4)$

$6y^7 + 4y^6 + 10y^5 - 8y^4$

c $(x + 2)^2 - 4$

$x^2 + 4x$

d $(x + 3)(x - 3) + 6x$

$x^2 - 9 + 6x$

e $(x + 4)(x + 3) + x - 4$

$x^2 + 8x + 8$

f $3x^2 + (2x - 1)(x - 2)$

$5x^2 - 5x + 2$

g $(2a + 3)(a - 5) - (a + 6)$

$2a^2 - 8a - 21$

h $x^2 + 2x - 3 + (2 + x)(2 + x)$

$2x^2 + 6x + 1$

i $x^2 - (x + 1)(x - 1)$

1

j $(3x - 2)(3x + 2) - (3x + 2)^2$

$-12x - 8$

k $(2x - 3)^2 + (2x + 3)^2$

$8x^2 + 18$

l $2(3x - 4)^2 - (3x - 4)(3x + 4)$

$9x^2 - 48x + 48$

m $(5x-1)^2-(5x+1)(5x-1)$

$2-10x$

n $2(x+y)^2-2(x-y)^2$

$8xy$

o $(x+y)^2-(x-y)^2+(x+y)(x-y)$

$x^2+4xy-y^2$

p $(2-x)^2-(2+x)^2$

$-8x$

q $(x^2+2)(x^2-2)$

x^4-4

r $(x^2+5)^2$

x^4+10x^2+25

s $(3ab-4c)(3ab+4c)$

$9a^2b^2-16c^2$

t $\left(x+\frac{2}{x}\right)^2$

$x^2+4+\frac{4}{x^2}$

u $\left(a-\frac{2}{a}\right)\left(a+\frac{2}{a}\right)$

$a^2-\frac{4}{a^2}$

v $[x+(y-2)]\,[x-(y-2)]$

x^2-y^2+4y-4

w $(a+3)^2-(a-3)^2$

$12a$

x $16-(z-4)(z+4)$

$32-z^2$

y $(x^2+y^2)-4x^2y^2$

$x^2+y^2-4x^2y^2$

z $(2a^2+5b^3)^2$

$4a^4+20a^2b^3+25b^6$

aa $(a+b)(a^2-ab+b^2)$

a^3+b^3

bb $(a-b)(a^2+ab+b^2)$

a^3-b^3

cc $[(x+3)(x-3)]^2$

x^4-18x^2+81

dd $\left[\left(\frac{x}{3}-5\right)\left(\frac{x}{3}+5\right)\right]^2$

$\frac{x^4}{81}-\frac{50x^2}{9}+625$

ee $(m+2)^2-(m+2)(m-2)-(m-2)^2$

$-m^2+8m+4$

ff $2(6-d)^2-(d+5)^2-(6-d)(d+5)$

$2d^2-35d+17$

Factorising Perfect Squares

Identify a perfect square trinomial

A **perfect square trinomial** is the expansion of a perfect square.

It is of the form $a^2 + 2ab + b^2$.

1. Find the square roots of the first term $\sqrt{a^2} = a$ and last term $\sqrt{b^2} = b$.
2. Check that the middle term is $2ab$.

✗	✓	✓	✓	✗
$x^2 + 6x + 16$	$x^2 + 6x + 9$	$x^2 - 10x + 25$	$4x^2 + 12x + 9$	$4x^2 + 10x + 9$

Decide if these are perfect square trinomials.

a $x^2 + 6x + 9$	**b** $x^2 + 7x + 9$	**c** $x^2 - 8x + 16$
d $x^2 - 8x + 9$	**e** $9x^2 - 12x + 4$	**f** $4x^2 + 20x + 25$
g $x^2 + 10x + 16$	**h** $x^2 - 14x + 49$	

Factorising Perfect Square Trinomials

$a^2 + 2ab + b^2$	$a^2 - 2ab + b^2$
$= (a + b)^2$	$= (a - b)^2$

1. Find the square roots of the first term $\sqrt{a^2} = a$ and last term $\sqrt{b^2} = b$.
2. Check whether the middle term is $2ab$.
3. Write $(a + b)^2$ or $(a - b)^2$ based on the sign of the middle term.

Example Factorise perfect square trinomials

$x^2 - 6x + 9$	$9x^2 + 30x + 25$
1. $\sqrt{x^2} = x$ $\quad \sqrt{9} = 3$	1. $\sqrt{9x^2} = 3x$ $\quad \sqrt{25} = 5$
2. Check: $2 \times x \times 3 = 6x$ ✓	2. Check: $2 \times 3x \times 5 = 30x$ ✓
3. $= (x - 3)^2$	3. $= (3x + 5)^2$

Guided Practice

Factorise these expressions

a $x^2 + 10x + 25$ $(x + 5)^2$	**b** $x^2 + 18x + 81$ $(x + 9)^2$
c $p^2 + 2pq + q^2$ $(p + q)^2$	**d** $x^2 - 16x + 64$ $(x - 8)^2$
e $25x^2 + 60x + 36$ $(5x + 6)^2$	**f** $x^6 - 14x^3 + 49$ $(x^3 - 7)^2$

Key Ideas

1. A perfect square is a t..... of the form $a^2 + 2ab + b^2$:
2. The first and third term are p..... s..... and the middle term is d..... the p..... of the square roots of the first and third term.

1 Factorise

a	$x^2 + 10x + 25$	**b**	$x^2 - 10x + 25$	**c**	$x^2 + 2x + 1$
	$(x + 5)^2$		$(x - 5)^2$		$(x + 1)^2$
d	$x^2 + 4x + 4$	**e**	$x^2 + 6x + 9$	**f**	$x^2 + 8x + 16$
	$(x + 2)^2$		$(x + 3)^2$		$(x + 4)^2$
a	$x^2 + 14x + 49$	**b**	$x^2 - 2x + 1$	**c**	$y^2 - 6y + 9$
	$(x + 7)^2$		$(x - 1)^2$		$(y - 3)^2$
d	$y^2 - 2y + 1$	**e**	$x^2 - 4x + 4$	**f**	$p^2 - 8p + 16$
	$(y - 1)^2$		$(x - 2)^2$		$(p - 4)^2$
g	$m^2 + 6m + 9$	**h**	$a^2 - 10a + 25$	**i**	$x^2 - 12x + 36$
	$(m + 3)^2$		$(a - 5)^2$		$(x - 6)^2$

2 Factorise

a	$x^2 + 6x + 9$	**b**	$4x^2 + 12x + 9$	**c**	$9x^2 + 12x + 4$
	$(x + 3)^2$		$(2x + 3)^2$		$(3x + 2)^2$
j	$9x^2 + 48x + 64$	**k**	$9x^2 - 30x + 25$	**l**	$4x^2 + 44x + 121$
	$(3x + 8)^2$		$(3x - 5)^2$		$(2x + 11)^2$
m	$49x^2 + 140x + 100$	**n**	$25x^2 - 20x + 4$	**o**	$9x^2 - 30x + 25$
	$(7x + 10)^2$		$(5x - 2)^2$		$(3x - 5)^2$
p	$m^2 - 2mn + n^2$	**q**	$r^2 + 2rt + t^2$	**r**	$9x^2 + 12xy + 16y^2$
	$(m - n)^2$		$(r + t)^2$		$(3x + 4y)^2$

3 Factorise

a	$x^2 + 2xy + y^2$	**b**	$x^2 - 2xy + y^2$	**c**	$4x^2 + 4xy + y^2$
	$(x + y)^2$		$(x - y)^2$		$(2x + y)^2$
d	$9x^2 - 6xy + y^2$	**e**	$25x^2 + 10xy + y^2$		
	$(3x - y)^2$		$(5x + y)^2$		

4 Factorise

a $2x^2 + 12x + 18$

$2(x + 3)^2$

b $3x^2 - 6x + 3$

$3(x - 1)^2$

c $5x^2 + 20x + 20$

$5(x + 2)^2$

d $4x^2 + 16xy + 16y^2$

$4(x + 2y)^2$

e $-2x^2 + 8x - 8$

$-2(x - 2)^2$

5 Factorise

a $x^4 + 6x^2 + 9$

$(x^2 + 3)^2$

b $x^4 - 10x^2 + 25$

$(x^2 - 5)^2$

c $x^6 + 4x^3 + 4$

$(x^3 + 2)^2$

d $4x^4 + 12x^2 + 9$

$(2x^2 + 3)^2$

e $x^4 + 2x^2y^2 + y^4$

$(x^2 + y^2)^2$

6 Factorise these perfect squares.

a $25a^4 - 10a^2 + 1$

$(5a^2 - 1)^2$

b $16x^4 + 24x^2 + 9$

$(4x^2 + 3)^2$

c $36 - 12x + x^2$

$(6 - x)^2$

d $49x^6 - 14x^3 + 1$

$(7x^3 - 1)^2$

e $81x^2 - 36xz + 4z^2$

$(9x - 2z)^2$

f $144m^6 - 72m^3 + 9$

$(12m^3 - 3)^2$

Factorising Difference of Two Squares

Identify a difference of two squares

A **difference of two squares** is of the form $a^2 - b^2$.

✗	✓	✓	✓	✗
$x^2 + 9^2$	$x^2 - 9$	$4x^2 - 25$	$9x^6 - 16y^4$	$x^3 - 9$

Decide if these are a difference of two squares.

a $x^2 - 16$	**b** $x^2 + 25$	**c** $4x^2 - 9$
d $x^2 - 12$	**e** $25a^2 - 49b^2$	**f** $16x^2 - 1$
g $x^5 - 16$	**h** $100x^{10} - 36y^8$	

Factorising Difference of Two Squares

$$a^2 - b^2$$
$$= (a - b)(a + b)$$

1. Find the square roots of each term $\sqrt{a^2} = a$ and $\sqrt{b^2} = b$.
2. Write $(a - b)(a + b)$.

Example Factorise difference of two squares

$a^2 - 36$

1. $\sqrt{a^2} = a$ $\quad \sqrt{36} = 6$
2. $(a - 6)(a + 6)$

$25x^6 - 9x^4$

1. $\sqrt{25x^6} = 5x^3$ $\quad \sqrt{9x^4} = 3x^2$
2. $(5x^3 - 3x^2)(5x^2 + 3x^2)$

Guided Practice

Factorise these expressions

a $x^2 - 1$

$(x - 1)(x + 1)$

b $25 - x^2$

$(5 - x)(5 + x)$

c $4x^2 - 9$

$(2x - 3)(2x + 3)$

d $49x^2 - 4y^2$

$(7x - 2y)(7x + 2y)$

e $a^2 - \frac{1}{9}$

$\left(a - \frac{1}{3}\right)\left(a + \frac{1}{3}\right)$

f $\frac{y^2}{16} - \frac{4}{9}$

$\left(\frac{y}{4} - \frac{2}{3}\right)\left(\frac{y}{4} + \frac{2}{3}\right)$

Key Ideas

1. If an expression involves a …… of two terms, both of which are …… ……, you factorise by using the rule: $a^2 - b^2 = (a - b)(a + b)$

1 Complete these factorisations.

a $x^2 - 16 = x^2 - 4^2$
$= (x + 4)(__ - __)$
$(x + 4)(x - 4)$

b $x^2 - 144 = x^2 - (__)^2$
$= (__ + 12)(x - __)$
$(x + 12)(x - 12)$

c $16x^2 - 1 = (__)^2 - (__)^2$
$= (4x + __)(__ - 1)$
$(4x + 1)(4x - 1)$

d $9a^2 - 4b^2 = (__)^2 - (__)^2$
$= (3a + __)(__ - 2b)$
$(9a + 2b)(9a - 2b)$

2 Factorise the following using difference of two squares.

a $x^2 - 9$
$(x - 3)(x + 3)$

b $y^2 - 25$
$(y - 5)(y + 5)$

c $y^2 - 1$
$(y - 1)(y + 1)$

d $x^2 - 16$
$(x - 4)(x + 4)$

e $a^2 - 81$
$(a - 9)(a + 9)$

f $x^2 - y^2$
$(x - y)(x + y)$

g $16 - a^2$
$(4 - a)(4 + a)$

h $25 - x^2$
$(5 - x)(5 + x)$

i $1 - b^2$
$(1 - b)(1 + b)$

j $36 - y^2$
$(6 - y)(6 + y)$

k $121 - b^2$
$(11 - b)(11 + b)$

l $x^2 - 400$
$(x - 20)(x + 20)$

3 Factorise

a $x^2 - 1$
$(x + 1)(x - 1)$

b $x^2 - 4$
$(x + 2)(x - 2)$

c $x^2 - 9$
$(x + 3)(x - 3)$

d $x^2 - 25$
$(x + 5)(x - 5)$

e $x^2 - 100$
$(x + 10)(x - 10)$

4 Factorise the following using difference of two squares.

a $4x^2 - 25$
$(2x - 5)(2x + 5)$

b $25b^2 - 4$
$(5b - 2)(5b + 2)$

c $100y^2 - 9$
$(10y - 3)(10y + 3)$

d $1 - 4x^2$
$(1 - 2x)(1 + 2x)$

e $16 - 9y^2$
$(4 - 3y)(4 + 3y)$

f $4x^2 - 25y^2$
$(2x - 5y)(2x + 5y)$

g $4p^2 - 25q^2$
$(2p - 5q)(2p + 5q)$

h $25a^2 - 49b^2$
$(5a - 7b)(5a + 7b)$

i $81m^2 - 4n^2$
$(9m - 2n)(9m + 2n)$

5 Factorise

a $4x^2 - 1$

$(2x + 1)(2x - 1)$

b $4x^2 - 9$

$(2x + 3)(2x - 3)$

c $4x^2 - 25$

$(2x + 5)(2x - 5)$

d $4x^2 - 49$

$(2x + 7)(2x - 7)$

6 Factorise

a $x^2 - 9$

$(x + 3)(x - 3)$

b $4x^2 - 9$

$(2x + 3)(2x - 3)$

c $9x^2 - 9$

$9(x + 1)(x - 1)$

d $25x^2 - 9$

$(5x + 3)(5x - 3)$

e $100x^2 - 9$

$(10x + 3)(10x - 3)$

7 Factorise

a $x^2 - y^2$

$(x + y)(x - y)$

b $4x^2 - y^2$

$(2x + y)(2x - y)$

c $x^2 - 4y^2$

$(x + 2y)(x - 2y)$

d $4x^2 - 9y^2$

$(2x + 3y)(2x - 3y)$

e $25x^2 - 16y^2$

$(5x + 4y)(5x - 4y)$

8 $-4 + 9x^2$ may not seem like a difference of two squares, but

$$-4 + 9x^2 = 9x^2 - 4$$
$$= (3x + 2)(3x - 2)$$

Use this idea to factorise these expressions.

a $-9 + x^2$

$(x - 3)(x + 3)$

b $-121 + 16x^2$

$(4x - 11)(4x + 11)$

c $-25a^2 + 4$

$(2 - 5a)(2 + 5a)$

d $-25a^2 + 4b^2$

$(2b - 5a)(2b + 5a)$

e $-36a^2b^2 + c^2$

$(c - 6ab)(c + 6ab)$

f $-16x^2 + y^2z^2$

$(yz - 4x)(yz + 4x)$

9 Some expressions with fractions can be simplified as a difference of two squares.

a $x^2 - \frac{1}{4}$

$\left(x - \frac{1}{2}\right)\left(x + \frac{1}{2}\right)$

b $x^2 - \frac{4}{25}$

$\left(x - \frac{2}{5}\right)\left(x + \frac{2}{5}\right)$

c $25x^2 - \frac{9}{16}$

$\left(5x - \frac{3}{4}\right)\left(5x + \frac{3}{4}\right)$

d $\frac{x^2}{9} - 1$

$\left(\frac{x}{3} - 1\right)\left(\frac{x}{3} + 1\right)$

e $\frac{a^2}{4} - \frac{b^2}{9}$

$\left(\frac{a}{2} - \frac{b}{3}\right)\left(\frac{a}{2} + \frac{b}{3}\right)$

f $\frac{5x^2}{9} - \frac{5}{4}$

$5\left(\frac{x}{3} - \frac{1}{2}\right)\left(\frac{x}{3} + \frac{1}{2}\right)$

10 Factorise each of the following by first taking out the common factor.

a $3x^2 - 108$

$= 3(x^2 - 36)$

$=$

$3(x-6)(x+6)$

b $6x^2 - 24$

$6(x-2)(x+2)$

c $10a^2 - 10$

$10(a-1)(a+1)$

d $4y^2 - 64$

$4(y-4)(y+4)$

e $98 - 2x^2$

$2(7-x)(7+x)$

f $32 - 8m^2$

$8(2-m)(2+m)$

g $5x^2y^2 - 5$

$5(xy-1)(xy+1)$

h $3 - 3x^2y^2$

$3(1-xy)(1+xy)$

i $63 - 7a^2b^2$

$7(3+ab)(3-ab)$

j $ax^2 - ay^2$

$a(x+y)(x-y)$

k $4x^3 - 16x$

$4x(x+2)(x-2)$

11 Factorise these expressions, if possible.

a $x^2 - 9x$

$x(x-9)$

b $x^2 + 9x$

$x(x+9)$

c $x^2 + 9$

can't simplify

d $x^2 - 9$

$(x-3)(x+3)$

e $x^2 - 8$

can't simplify

f $2x^2 - 50$

$2(x-5)(x+5)$

g $50 - 2x^2$

$2(5-x)(5+x)$

h $200 - 2x^2$

$2(10-x)(10+x)$

i $300 - 3x^2$

$3(10-x)(10+x)$

j $300 - 27x^2$

$3(10-3x)(10+3x)$

k $12x^2 - 3y^2$

$3(2x-y)(2x+y)$

l $3x^2 - 3y^2$

$3(x-y)(x+y)$

m $3x^2 - 27y^2$

$3(x-3y)(x+3y)$

n $x^2 - 27y^2$

can't simplify

12 Olivia factorises $16x^2 - 4$ to get $(4x + 2)(4x - 2)$ but the answers say $4(2x + 1)(2x - 1)$.

a What should Olivia do to get from her answer to the actual answer?

Factorise 2 out of both sets of brackets.

$2(2x + 1)2(2x - 1) = 4(2x + 1)(2x - 1)$

b What should Olivia have done initially to avoid this issue?

Factorise out the HCF: 4

13 Amy factorises $25 - x^2$ to get $(5 + x)(5 - x)$ but the answers say $-(x + 5)(x - 5)$.

Are these the same? Justify your answer.

Yes. You can check by expanding both expressions, or notice that we factorised -1 out of the second set of brackets.

14 Some expressions can be factorised as a difference of two squares using surds.

a $x^2 - 7$ $= (x + \sqrt{7})(x - \sqrt{7})$	**b** $x^2 - 5$ $(x - \sqrt{5})(x + \sqrt{5})$	**c** $x^2 - 19$ $(x - \sqrt{19})(x + \sqrt{19})$
d $3x^2 - 4$ $(\sqrt{3}x - 2)(\sqrt{3}x + 2)$	**e** $5x^2 - 9$ $(\sqrt{5}x - 3)(\sqrt{5}x + 3)$	**f** $-9 + 2x^2$ $(\sqrt{2}x - 3)(\sqrt{2}x + 3)$

15 Factorise using surds, then simplify the surds.

a $x^2 - 8$ $(x - 2\sqrt{2})(x + 2\sqrt{2})$	**b** $x^2 - 45$ $(x - 3\sqrt{5})(x + 3\sqrt{5})$	**c** $x^2 - 20$ $(x - 2\sqrt{5})(x + 2\sqrt{5})$
d $x^2 - 48$ $(x - 4\sqrt{3})(x + 4\sqrt{3})$	**e** $x^2 - 50$ $(x - 5\sqrt{2})(x + 5\sqrt{2})$	**f** $x^2 - 200$ $(x - 10\sqrt{2})(x + 10\sqrt{2})$

16 Factorise fully by first factoring a common factor, then simplifying any surds.

a $5x^2 - 120$ $5(x - 2\sqrt{6})(x + 2\sqrt{6})$	**b** $3x^2 - 162$ $3(x - 3\sqrt{6})(x + 3\sqrt{6})$	**c** $7x^2 - 126$ $7(x - 3\sqrt{2})(x + 3\sqrt{2})$

17 Read the example for factorising $25 - (x+2)^2$

Worked Solution	Explanation
$25 - (x+2)^2$ $= \big(5 + (x+2)\big)\big(5 - (x+2)\big)$ $= (7+x)(3-x)$	I recognise $25 = 5^2$ and $(x+2)^2 = (x+2)^2$. So $a =$ ..... and $b =$ .....

18 Factorise the following.

a $(x+5)^2 - 9$ $(x+2)(x+8)$

b $(x+10)^2 - 16$ $(x+6)(x+14)$

c $(x-7)^2 - 1$ $(x-8)(x-6)$

d $49 - (x-1)^2$ $(8-x)(x+6)$

e $9 - (x+3)^2$ $-x(x+6)$

f $16 - (x+4)^2$ $-x(x+8)$

g $25 - (x+2)^2$ $(3-x)(x+7)$

h $100 - (x+4)^2$ $(6-x)(x+14)$

i $25 - (x-5)^2$ $x(10-x)$

j $49 - (x+3)^2$ $(4-x)(x+10)$

k $4 - (x+2)^2$ $-x(x+4)$

l $81 - (x+8)^2$ $(1-x)(x+17)$

m $(3x+5)^2 - x^2$ $(2x+5)(4x+5)$

n $(2y+7)^2 - y^2$ $(y+7)(3y+7)$

o $(3x-5y)^2 - 25y^2$ $3x(3x-10y)$

p $(x-6)^2 - 15$ $(x-6-\sqrt{15})(x-6+\sqrt{15})$

q $21 - (x+4)^2$ $(\sqrt{21}-x-4)(\sqrt{21}+x+4)$

r $19 - (x-1)^2$ $(\sqrt{19}-x+1)(\sqrt{19}+x-1)$

19 Richard factorised this expression but made a mistake.

$$9 - (x-1)^2 = (3+x-1)(3-x-1)$$
$$= (2+x)(2-x)$$

a Identify the mistake.

He should have kept the brackets: $(3+x-1)(3-(x-1))$

b Write the correct factorisation.

$(2+x)(4-x)$

20 Factorise

a $x^2 - 9$

$(x + 3)(x - 3)$

b $x^4 - 9$

$(x^2 + 3)(x^2 - 3)$

c $x^6 - 9$

$(x^3 + 3)(x^3 - 3)$

d $x^{2n} - 9$

$(x^n + 3)(x^n - 3)$

21 Factorise

a $x^4 - 25$

$(x^2 + 5)(x^2 - 5)$

b $4x^4 - 25$

$(2x^2 + 5)(2x^2 - 5)$

c $x^4 - 25y^2$

$(x^2 + 5y)(x^2 - 5y)$

d $4x^4 - 25y^6$

$(2x^2 + 5y^3)(2x^2 - 5y^3)$

e $9x^6 - 4y^4$

$(3x^3 + 2y^2)(3x^3 - 2y^2)$

22 Some expressions with powers of 4 can be factorised as a difference of two squares. Look at the example and factorise the following.

$$x^4 - y^4 = (x^2 + y^2)(x^2 - y^2)$$
$$= (x^2 + y^2)(x + y)(x - y)$$

a $1 - w^4$

$(1 - w)(1 + w)(w^2 + 1)$

b $\frac{x^4}{3} - \frac{y^4}{3}$

$\frac{(x - y)(x + y)(x^2 + y^2)}{3}$

c $2a^4 - 32b^4$

$2(a - 2b)(a + 2b)(a^2 + 4b^2)$

d $\frac{3a^8}{16} - \frac{3b^4}{625}$

$3\left(\frac{a^2}{2} - \frac{b}{5}\right)\left(\frac{a^2}{2} + \frac{b}{5}\right)\left(\frac{a^4}{4} + \frac{b^2}{25}\right)$

e $5w^5 - 80w$

$5w(w - 2)(w + 2)(w^2 + 4)$

f $w^6 - w^4 - w^2 + 1$

$(w - 1)^2(w + 1)^2(w^2 + 1)$

Factorising by Noticing Binomial Factors

Review

Divide terms

a	$3a \div a$	**b**	$3xy \div xy$	**c**	$3xy \div 3y$
d	$3(x+y) \div 3$	**e**	$3(x+y) \div (x+y)$	**f**	$3x(a+b) \div (a+b)$
g	$5x(2a+3b) \div (2a+3b)$	**h**	$-2x(p-q) \div (p-q)$	**i**	$4(x+y)^2 \div (x+y)$

Factorising by Noticing Binomial Common Factors

$$a(c+d) + b(c+d)$$
$$= (c+d)(a+b)$$

1. Identify the **binomial** that divides into all terms. Write it outside a second set of brackets.
2. Divide each term by the binomial HCF. Write the results inside a second set of brackets.

Example Factorise by noticing the binomial HCF

$3(x+y) + x(x+y)$	$5(2x-y) - 3x(2x-y)$
1. $(x+y)(\quad)$	1. $(2x-y)(\quad)$
2. $(x+y)(3+x)$	2. $(2x-y)(5-3x)$

Guided Practice

Factorise.

a $4(x+3) + x(x+3)$

$= (x+3)(\quad)$

$(x+3)(4+x)$

b $3(x+1) - x(x+1)$

$(x+1)(3-x)$

c $x(x-7) + 2(x-7)$

$(x-7)(x+2)$

d $8(a-4) - a(a-4)$

$(a-4)(8-a)$

e $x(7+3x) + (7+3x)$

$(7+3x)(x+1)$

f $y(2x-5) - (2x-5)$

$(2x-5)(y-1)$

Factorising by Noticing Binomial Common Factors

Sometimes the binomial common factor is not obvious.

1. Factorise a pair of terms such that the expression is inside brackets.
2. Factorise by noticing the binomial HCF.

Example Factorise by noticing binomial HCF after factorising one pair of terms

$7 - 2x + x(7 - 2x)$

1. $1(7 - 2x) + x(7 - 2x)$
2. $(7 - 2x)(1 + x)$

$7x(x - 4) - x + 4$

1. $7x(x - 4) - (x - 4)$
2. $(x - 4)(7x - 1)$

Guided Practice

Factorise these expressions.

a $7 + 3x + x(7 + 3x)$

$= 1(7 + 3x) + x(7 + 3x)$

$=$

$(7 + 3x)(1 + x)$

b $y(4y - 1) - (4y - 1)$

$(4y - 1)(y - 1)$

c $y(2x - 5) - 2x + 5$

$(2x - 5)(y - 1)$

d $2x(x + 1) - x - 1$

$(x + 1)(2x - 1)$

e $5y(x - 7) - x + 7$

$(x - 7)(5y - 1)$

f $12x^2(y - 3) + 3 - y$

$(y - 3)(12x^2 - 1)$

Key Ideas

1. We can factorise an expression in the form $a(c + d) + b(c + d)$ by taking out the common factor, $(c + d)$, then writing ……
2. Sometimes, you may need to factor out 1 or …… for a pair of terms to see the binomial common factor.

1 Fill in the missing information to finish factorising.

a $2(x+1)+a(x+1)=(x+1)(_____)$

$2+a$

b $3(x+3)-a(x+3)=(x+3)(_____)$

$3-a$

c $5(x+5)-a(x+5)=(x+5)(_____)$

$5-a$

d $a(x+7)+4(x+7)=(x+7)(_____)$

$a+4$

e $a(x-3)+(x-3)=(x-3)(_____)$

$a+1$

f $a(x+4)-(x+4)=(x+4)(_____)$

$a-1$

g $(x-3)-a(x-3)=(x-3)(_____)$

$1-a$

h $(4-x)+2a(4-x)=(4-x)(_____)$

$1+2a$

2 Factorise:

a $1(x+3)+x(x+3)$

$(x+1)(x+3)$

b $2(x+3)+x(x+3)$

$(x+2)(x+3)$

c $5(x+3)+x(x+3)$

$(x+5)(x+3)$

d $x(x+3)+x(x+3)$

$2x(x+3)$

3 Factorise:

a $5(x+3)+x(x+3)$

$(x+5)(x+3)$

b $5(x+3)-x(x+3)$

$(5-x)(x+3)$

c $-5(x+3)+x(x+3)$

$(x-5)(x+3)$

d $-5(x+3)-x(x+3)$

$-(x+5)(x+3)$

4 Factorise the following expressions by noticing the binomial factor.

a $4(x+3)+x(x+3)$

$(x+3)(4+x)$

b $3(x+1)+x(x+1)$

$(x+1)(3+x)$

c $7(m-3)+m(m-3)$

$(m-3)(7+m)$

d $x(x-7)+2(x-7)$

$(x-7)(x+2)$

e $8(a+4)-a(a+4)$

$(a+4)(8-a)$

f $y(y+3)-2(y+3)$

$(y+3)(y-2)$

g $a(x+2)+(x+2)$

$(x+2)(a+1)$

h $t(2t+5)+3(2t+5)$

$(2t+5)(t+3)$

i $y(4y-1)-(4y-1)$

$(4y-1)(y-1)$

5 Factorise these expressions.

a $x(x+1)+3(x+1)$

$(x+1)(x+3)$

b $x(x+1)+2(x+1)$

$(x+1)(x+2)$

c $x(x+1)+1(x+1)$

$(x+1)^2$

d $3x(x+1)+(x+1)$

$(x+1)(3x+1)$

e $3x(x+1)-(x+1)$

$(x+1)(3x-1)$

f $3x(2x+1)-(2x+1)$

$(2x+1)(3x-1)$

g $3x(1+2x)-(2x+1)$

$(1+2x)(3x-1)$

h $3x(1-2x)-(1-2x)$

$(1-2x)(3x-1)$

i $2x(1-2x)-(1-2x)$

$(1-2x)(2x-1)$ or $-(2x-1)^2$

j $2x(1-2x)-(2x-1)$

$(1-2x)(2x+1)$

6 Using the fact that $a-b=-(b-a)$, you can factorise $x(x-2)-5(2-x)$ by following these steps:

$$x(x-2)-5(2-x)=x(x-2)+5(x-2)$$
$$=(x-2)(x+5)$$

Use this idea to factorise these expressions.

a $x(x-4)+3(4-x)$

$=x(x-4)-3(x-4)$

$=$

$(x-4)(x-3)$

b $x(x-5)-2(5-x)$

$(x-5)(x+2)$

c $x(x-3)-3(3-x)$

$(x-3)(x+3)$

d $3x(x-4)+5(4-x)$

$(x-4)(3x-5)$

e $3(2x-5)+x(5-2x)$

$(2x-5)(3-x)$

f $2x(x-2)+(2-x)$

$(x-2)(2x-1)$

g $-4(3-x)-x(x-3)$

$(x-3)(4-x)$

h $x(x-5)+(10-2x)$

$(x-2)(x-5)$

i $x(x-3)+(6-2x)$

$(x-2)(x-3)$

7 Factorise:

a $5(x-4)+3(4-x)$

$2(x-4)$

b $x(x-4)-3(4-x)$

$(x+3)(x-4)$

c $2x(3-x)+5(x-3)$

$-(2x+5)(x-3)$

d $x^2(2-x)-4(x-2)$

$(x^2+4)(2-x)$

8 Factorise these expressions using an appropriate method.

a $6a+30$

$6(a+5)$

b $5x-15$

$5(x-3)$

c $8b+18$

$2(4b+9)$

d x^2-4x

$x(x-4)$

e y^2+9y

$y(y+9)$

f a^2-3a

$a(a-3)$

g $x^2y-4xy+xy^2$

$xy(x-4+y)$

h $6ab-10a^2b+8ab^2$

$2ab(3-5a+4b)$

i $m(m+5)+2(m+5)$

$(m+5)(m+2)$

j $x(x+3)-2(x+3)$

$(x+3)(x-2)$

k $x(2x-1)+(-2x+1)$

$(2x-1)(x-1)$

l $y(3-2y)-5(2y-3)$

$(3-2y)(y+5)$

9 Factorise fully:

a $3(x+2)+x(x+2)$

$(x+3)(x+2)$

b $x(x+2)+x^2(x+2)$

$x(x+1)(x+2)$

c $3x^2(x+2)+5(x+2)$

$(3x^2+5)(x+2)$

d $x^3(x+2)-4(x+2)$

$(x^3-4)(x+2)$

e $(x+2)^2(x+1)+3(x+2)$

$(x+2)(x^2+3x+5)$

Factorising by Grouping in Pairs

ⓘ Factorise by Grouping in Pairs

$$ac + ad + bc + bd$$

Factorise expressions with **four terms** with no common factor using **grouping in pairs**:

1. Pair the terms.
2. Factorise each pair separately.
3. Factorise using the binomial common factor.

⚠ You may need to take out a negative factor to get a binomial common factor.

Example Factorise by grouping in pairs

$6xy + 9x + 4y + 6$

1. $6xy + 9x + 4y + 6$
2. $3x(2y + 3) + 2(2y + 3)$
3. $(2y + 3)(3x + 2)$

The first step is to ..... up the first two terms and the last two terms.

From the first pair, factor out .....

From the second pair, factor out .....

The binomial common factor is .....

Guided Practice

Factorise:

a $2x + 8 + bx + 4b$

$(x + 4)(2 + b)$

b $m^2 - 2m + 3m - 6$

$(m - 2)(m + 3)$

c $3x^2 + 6x - 2x - 4$

$(x + 2)(3x - 2)$

d $2ax + bx - 2ay - by$

$(2a + b)(x - y)$

Example Factorise by grouping in pairs

$2x - 4 + 6y - 3xy$

1. $2x - 4 + 6y - 3xy$
2. $2(x - 2) + 3y(2 - x)$
3. $2(x - 2) - 3y(x - 2)$

$(x - 2)(2 - 3y)$

In this example, we needed to factor out a ..... from $3y(2 - x)$ to make ..... so that we could get a c..... f.....

Guided Practice

Factorise:

a $6x - 3 + 5y - 10xy$

$= 3(2x - 1) + 5y(1 - 2x)$

$=$

$(2x - 1)(3 - 5y)$

b $3x - 15 + 5y - xy$

$(x - 5)(3 - y)$

c $6 - 2a + a^3 - 3a^2$

$(3 - a)(2 - a^2)$

d $2ax - bx + by - 2ay$

$(2a - b)(x - y)$

Key Ideas

1. If there are four terms that share no common factor, factorise by ..... in .....
2. Sometimes you may need to factor out a ..... to be able to get a binomial common factor.

1 Fill in the missing information to finish factorising.

a

$x^2 + 3x + 4x + 12 = (x^2 + 3x) + (_____)$

$= __(x + 3) + __(x + 3)$

$= (x + __)\,(x + __)$

$(x + 3)(x + 4)$

b

$x^2 + 3x - 4x - 12 = (x^2 + 3x) - (_____)$

$= x(__ + __) - 4(__ + __)$

$= (_____)(_____)$

$(x + 3)(x - 4)$

2 Factorise the following expressions by grouping in pairs.

a $x^2 + 3x + 2x + 6$

$= x(x + 3) + 2(x + 3)$

$=$

$(x + 3)(x + 2)$

b $x^2 + 4x + 3x + 12$

$(x + 4)(x + 3)$

c $x^2 + 7x + 2x + 14$

$(x + 7)(x + 2)$

d $x^2 - 6x + 4x - 24$

$(x - 6)(x + 4)$

e $x^2 - 4x + 6x - 24$

$(x - 4)(x + 6)$

f $x^2 - 3x + 10x - 30$

$(x - 3)(x + 10)$

g $x^2 + 2x - 18x - 36$

$(x + 2)(x - 18)$

h $x^2 + 3x - 14x - 42$

$(x + 3)(x - 14)$

i $x^2 - 3x - 3xc + 9c$

$(x - 3)(x - 3c)$

j $x^2 - 2x - xa + 2a$

$(x - 2)(x - a)$

k $x^2 - 5x - 3xa + 15a$

$(x - 5)(x - 3a)$

l $x^2 + 4x - 18x - 72$

$(x + 4)(x - 18)$

3 Factorise:

a $x^2 + 4x + 3x + 12$

$(x + 3)(x + 4)$

b $x^2 + 4x - 3x - 12$

$(x - 3)(x + 4)$

c $x^2 - 4x + 3x - 12$

$(x + 3)(x - 4)$

d $x^2 - 4x - 3x + 12$

$(x - 3)(x - 4)$

4 Factorise these expressions.

a $3ab + 5bc + 3ad + 5cd$

$(3a + 5c)(b + d)$

b $4ab - 7ac + 4bd - 7cd$

$(4b - 7)(a + d)$

c $2xy - 8xz + 3wy - 12wz$

$(3w + 2x)(y - 4z)$

d $5rs - 10r + st - 2t$

$(s - 2)(5r + t)$

e $4x^2 + 12xy - 3x - 9y$

$(4x - 3)(x + 3y)$

f $2ab - a^2 - 2bc + ac$

$(a - 2b)(c - a)$

5 Factorise these expressions. Remember to use a factor of 1 where necessary, for example,

$$x^2 - 3x + x - 3 = x(x - 3) + 1(x - 3)$$

a $x^2 - bx + x - b$

$(x + 1)(x - b)$

b $x^2 - cx + x - c$

$(x + 1)(x - c)$

c $x^2 + bx + x + b$

$(x + 1)(b + x)$

d $x^2 + cx - x - c$

$(x - 1)(c + x)$

e $x^2 + ax - x - a$

$(x - 1)(x + a)$

f $x^2 - bx - x + b$

$(x - 1)(x - b)$

6 Factorise:

a $2x^2 + 3x + 4x + 6$

$(x + 2)(2x + 3)$

b $2x^2 + 4x + 3x + 6$

$(x + 2)(2x + 3)$

c $2x^2 - 3x + 4x - 6$

$(x + 2)(2x - 3)$

d $2x^2 - 4x - 3x + 6$

$(2x - 3)(x - 2)$

7 Factorise these expressions.

a $x^2 + 2x + 3x + 6$

$(x + 2)(x + 3)$

b $x^2 + 3x + 2x + 6$

$(x + 3)(x + 2)$

c $3x^2 + 3x + 2x + 2$

$(x + 1)(3x + 2)$

d $3x^2 + 3x + 4x + 4$

$(x + 1)(3x + 4)$

e $3x^2 - 3x + 4x - 4$

$(x - 1)(3x + 4)$

f $3x^2 - 3x - 4x + 4$

$(x - 1)(3x - 4)$

g $3x^2 + 3x - 4x - 4$

$(x + 1)(3x - 4)$

h $3x^2 - 4x + 3x - 4$

$(3x - 4)(x + 1)$

i $5x^2 - 4x + 5x - 4$

$(5x - 4)(x + 1)$

j $-5x^2 + 4x - 5x + 4$

$-(5x - 4)(x + 1)$

8 The expression $xa - 21 + 7a - 3x$ currently do not have common factors between pairs.
You can rearrange the expression in two ways and factorise. You should get the same answer.

Method 1

$xa + 7a - 3x - 21 = a(x + 7) - 3(_____)$

$= _____$

Method 2

$xa - 3x + 7a - 21 = x(a - 3) + 7(_____)$

$= _____$

Complete the factorisation of the following by rearranging in two different ways.

a $xb - 6 - 3b + 2x$

$xb - 3b - 6 + 2x$

$xb + 2x - 6 - 3b$

$(b + 2)(x - 3)$

b $4m^2 - 15n + 6m - 10mn$

$(2m + 3)(2m - 5n)$

c $4a - 6b^2 + 3b - 8ab$

$(2b - 1)(-4a - 3b)$

9 Factorise these expressions by first rearranging the terms.

a $2x^2 - 7 - 14x + x$

$= 2x^2 - 14x - 7 + x$

$=$

$(2x + 1)(x - 7)$

b $5x + 2x + x^2 + 10$

$(x + 5)(x + 2)$

c $2x^2 - 3 - x + 6x$

$(2x - 1)(x + 3)$

d $3x - 8x - 6x^2 + 4$

$(1 - 2x)(3x + 4)$

e $11x - 5a - 55 + ax$

$(a + 11)(x - 5)$

f $12y + 2x - 8xy - 3$

$(2x - 3)(1 - 4y)$

g $6m - n + 3mn - 2$

$(3m - 1)(n + 2)$

h $15p - 8r - 5pr + 24$

$(5p + 8)(3 - r)$

i $16x - 3y - 8xy + 6$

$(8x + 3)(2 - y)$

10 These expressions are not fully factorised. Finish factorising by taking out a common factor.

a $(2x+4)(x+5)$

$= 2(x+2)(x+5)$

b $(x+2)(2x+10)$

$2(x+2)(x+5)$

c $(3x+6)(2x+11)$

$3(x+2)(2x+11)$

d $(9x+6)(2x+11)$

$3(3x+2)(2x+11)$

e $(9x+6)(2x+10)$

$6(3x+2)(x+5)$

f $(9x+6)(5x+10)$

$15(3x+2)(x+2)$

g $(9x+6)(15x+10)$

$15(3x+2)^2$

h $2(9x+6)(15x+10)$

$30(3x+2)^2$

i $2(6x-9x^2)(15x+10)$

$30x(2-3x)(3x+2)$

j $2x^3(6x-9x^2)(15x+10)$

$30x^4(2-3x)(3x+2)$

11 Factorise:

a x^3+x^2-x-1

$(x+1)^2(x-1)$

b x^3-x^2-x+1

$(x-1)^2(x+1)$

c $2x^3+6x^2-2x-6$

$2(x+1)(x-1)(x+3)$

12 Factorise by grouping.

a $2(a-3)-x(a-3)-c(a-3)$

$(a-3)(2-x-c)$

b $b(2a+1)+5(2a+1)-a(2a+1)$

$(2a+1)(b+5-a)$

c $x(a+1)-4(a+1)-ba-b$

$(a+1)(x-4-b)$

d $c(1-a)-x+ax+2-2a$

$(1-a)(c-x+2)$

e $a^2-3ac-2ab+6bc+3abc-9bc^2$

$(a-3c)(a-2b+3bc)$

f $8z-4y+3x^2+xy-12x-2xz$

$(x-4)(3x+y-2z)$

13 Factorise by noticing binomial factors, then simplify.
This skill will be helpful when applying chain rule and product rule in Year 11 calculus.

a $3x(5x+2)^2 - 2(5x+2)$

$= (5x+2)(3x(5x+2) - 2)$

$= (5x+2)(15x^2 + 6x - 2)$

b $3x(7x-1)^2 - 2(7x-1)$

$= (7x-1)($ ..... $)$

$=$

$(7x-1)(21x^2 - 3x - 2)$

c $(x+3)^2 + 5(x+3)$

$(x+3)(x+8)$

d $(y+1)^2 - 4(y+1)$

$(y+1)(y-3)$

e $25x(2x+1) + 15(2x+1)$

$5(2x+1)(5x+3)$

f $25x(2x+1) - 15(2x+1)$

$5(2x+1)(5x-3)$

g $25x(2x+1)^2 + 15(2x+1)$

$5(2x+1)(10x^2+5x+3)$

h $25x(2x+1) + 15(2x+1)^2$

$5(2x+1)(11x+3)$

i $50x(2x+1) - 15(2x+1)^2$

$5(2x+1)(4x-3)$

j $25x(2x+1)^3 - 15(2x+1)^2$

$5(2x+1)^2(10x^2+5x-3)$

k $5ax(2x+1)^3 - 3a(2x+1)^2$

$a(2x+1)^2(10x^2+5x-3)$

l $5x\sqrt{2x+1} - 3\sqrt{2x+1}$

$\sqrt{2x+1}(5x-3)$

m $5x(2x+1)(5x-2) - 3(2x+1)^2(5x-2)^2$

$-2(5x-2)(2x+1)(15x^2-x-3)$

n $5x(2x-1)^2 - 3x(1-2x)$

$2x(5x-1)(2x-1)$

Factorising Non-Monic Quadratic Trinomials

ⓘ Factorising Non-Monic Quadratic Trinomials

Non-monic means the coefficient of x^2 is not 1.
Non-monic quadratic trinomials are of the form $ax^2 + bx + c$.

To factorise a non-monic quadratic trinomial: $ax^2 + bx + c = 0$

1. Identify the product: ac
2. Identify the sum: b
3. Find a factor pair, α and β, that has product of ac and sum to b.
4. Split bx to $\alpha x + \beta x$
5. Factorise by grouping in pairs.

Example Factorise non-monic quadratic trinomials

$$5x^2 - 13x + 6$$

1. P: $5 \times 6 = 30$
2. S: -13
3. F: $10, -3$
4. Split $= 5x^2 - 10x - 3x + 6$
5. Factorise $= 5x(x - 2) - 3(x - 2)$
   $= (x - 2)(5x - 3)$

Guided Practice

Factorise:

a $3x^2 + 4x + 1$

P: 3
S: 4
F: .....

$(3x + 1)(x + 1)$

b $2x^2 + 11x + 5$

P: .....
S: .....
F: .....

$(2x + 1)(x + 5)$

c $6x^2 + 5x - 6$

P: .....
S: .....
F: .....

$(3x - 2)(2x + 3)$

d $2x^2 - 7x - 15$

P: .....
S: .....
F: .....

$(2x + 3)(x - 5)$

Key Ideas

1. Non-monic quadratic means the coefficient of …… is not ……
2. To factorise non-monic quadratic t……, split the …… term, then factorise by ……

1 Factorise these expressions.

a $6x^2 + 13x + 6$

P: .....
S: .....
F: .....

$(3x + 2)(2x + 3)$

b $3x^2 - 11x - 4$

P: .....
S: .....
F: .....

$(3x + 1)(x - 4)$

c $12x^2 + x - 6$

P: .....
S: .....
F: .....

$(3x - 2)(4x + 3)$

d $10x^2 - 11x - 6$

P: .....
S: .....
F: .....

$(5x + 2)(2x - 3)$

e $21x^2 - 20x + 4$

P: .....
S: .....
F: .....

$(7x - 2)(3x - 2)$

f $15x^2 - 13x + 2$

P: .....
S: .....
F: .....

$(5x - 1)(3x - 2)$

g $13x^2 - 7x - 6$

$(x + 1)(2x + 3)$

h $2x^2 + 5x - 12$

$(2x - 3)(x + 4)$

i $6x^2 - 13x + 6$

$(3x - 2)(2x - 3)$

j $9x^2 + 9x - 10$

$(3x - 2)(3x + 5)$

2 Factorise $4x^2 - 4x - 15$ in two different ways to show that the order you split the x term doesn't change the answer.

a

$$4x^2 - 4x - 15 = 4x^2 - 10x + 6x - 15$$
$$= 2x(____) + 3(____)$$
$$= (2x - 5)(____)$$

$(2x - 5)(2x + 3)$

b

$$4x^2 - 4x - 15 = 4x^2 + 6x - 10x - 15$$
$$= 2x(____) - 5(____)$$
$$= (____)(____)$$

$(2x - 5)(2x + 3)$

3 Factorise these expressions.

a $2x^2 + 3x + 1$

$(2x + 1)(x + 1)$

b $2x^2 - 3x + 1$

$(2x - 1)(x - 1)$

c $2x^2 - 3x - 2$

$(2x + 1)(x - 2)$

d $2x^2 + 3x - 2$

$(2x - 1)(x + 2)$

e $2x^2 + 5x + 2$

$(2x + 1)(x + 2)$

f $2x^2 - 5x + 3$

$(2x - 3)(x - 1)$

g $2x^2 + 5x + 3$

$(2x + 1)(x + 1)$

h $2x^2 - x - 3$

$(2x - 3)(x + 1)$

i $2x^2 + x - 3$

$(2x + 3)(x - 1)$

j $2x^2 + x - 1$

$(2x - 1)(x + 1)$

k $2x^2 - x - 1$

$(2x + 1)(x - 1)$

l $4x^2 - 1$

$(2x - 1)(2x + 1)$

m $4x^2 + 4x + 1$

$(2x + 1)^2$

n $4x^2 - 4x + 1$

$(2x - 1)^2$

o $4x^2 + 5x + 1$

$(4x + 1)(x + 1)$

p $4x^2 - 5x + 1$

$(4x - 1)(x - 1)$

q $4x^2 + 3x - 1$

$(4x - 1)(x + 1)$

r $4x^2 - 3x - 1$

$(4x + 1)(x - 1)$

4 Factorise:

a $2x^2 + 7x + 3$

$(2x + 1)(x + 3)$

b $2x^2 - 7x + 3$

$(2x - 1)(x - 3)$

c $2x^2 + 5x - 3$

$(2x - 1)(x + 3)$

d $2x^2 - 5x - 3$

$(2x + 1)(x - 3)$

5 Factorise:

a $x^2 + 5x + 4$

$(x + 1)(x + 4)$

b $2x^2 + 9x + 4$

$(2x + 1)(x + 4)$

c $3x^2 + 13x + 4$

$(3x + 1)(x + 4)$

d $4x^2 + 17x + 4$

$(4x + 1)(x + 4)$

6 Factorise:

a $2x^2 + 3x + 1$

$(2x + 1)(x + 1)$

b $2x^2 + 5x + 2$

$(2x + 1)(x + 2)$

c $2x^2 + 7x + 3$

$(2x + 1)(x + 3)$

d $2x^2 + 9x + 4$

$(2x + 1)(x + 4)$

7 Factorise by first taking out the common factor.

a $6x^2 + 38x + 40$

$2(3x + 4)(x + 5)$

b $6x^2 - 15x - 36$

$3(2x + 3)(x - 4)$

c $48x^2 - 18x - 3$

$3(8x + 1)(2x - 1)$

d $32x^2 - 88x + 60$

$4(4x - 5)(2x - 3)$

e $16x^2 - 24x + 8$

$8(2x - 1)(x - 1)$

f $90x^2 + 90x - 100$

$10(3x - 2)(3x + 5)$

g $-50x^2 - 115x - 60$

$-5(5x + 4)(2x + 3)$

h $12x^2 - 36x + 27$

$3(2x - 3)^2$

8 Factorise the following.

a $18x^2 + 27x + 10$

$(6x + 5)(3x + 2)$

b $21x^2 + 22x - 8$

$(7x - 2)(3x + 4)$

c $40x^2 - x - 6$

$(8x + 3)(5x - 2)$

d $45x^2 - 46x + 8$

$(9x - 2)(5x - 4)$

9 Factorise:

a $-x^2 - 5x - 6$

$-(x + 2)(x + 3)$

b $-2x^2 - 5x - 3$

$-(2x + 3)(x + 1)$

c $-2x^2 + 5x - 3$

$-(2x - 3)(x - 1)$

d $-3x^2 + x + 2$

$-(3x + 2)(x - 1)$

e $-6x^2 + x + 2$

$-(3x - 2)(2x + 1)$

10 Show that $-12x^2 - 5x + 3$ factorises to $(1 - 3x)(4x + 3)$.

P: -36, S: -5, F: $-9,4$

$-12x^2 - 9x + 4x + 3$

$= -3x(4x - 3) + (4x - 3)$

$= (4x - 3)(1 - 3x)$

Hence factorise these expressions.

a $-8x^2 + 2x + 15$

$(3 - 2x)(4x + 5)$

b $-6x^2 + 11x + 10$

$(5 - 2x)(3x + 2)$

c $-12x^2 + 13x + 4$

$(4 - 3x)(4x + 1)$

d $-8x^2 + 18x - 9$

$(3 - 4x)(2x - 3)$

e $-14x^2 + 39x - 10$

$(2 - 7x)(2x - 5)$

f $-15x^2 - x + 6$

$(3 - 5x)(3x + 2)$

11 Factorise:

a $2x^2 + 3xy + y^2$

$(2x + y)(x + y)$

b $2x^2 - xy - y^2$

$(2x + y)(x - y)$

c $3x^2 + 5xy + 2y^2$

$(3x + 2y)(x + y)$

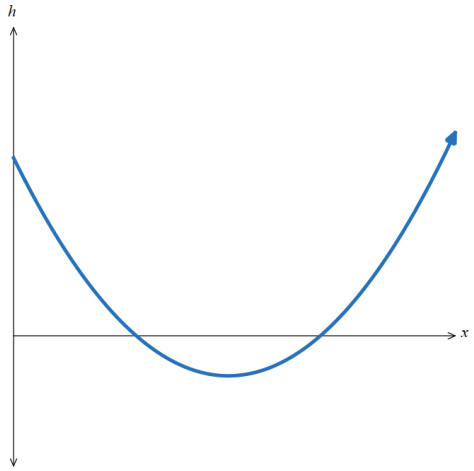
d $6x^2 - xy - y^2$

$(3x + y)(2x - y)$

e $6x^2 + 5xy - 6y^2$

$(3x - 2y)(2x + 3y)$

12 A cable is suspended across a farm channel. The height (h), in metres, of the cable above the water surface is modelled by the equation $h = 3x^2 - 21x + 30$, where x metres is the distance from one side of the channel. The graph of h vs x is shown below.



a Factorise the right-hand side of the equation.

$3(x - 2)(x - 5)$

b Determine the height of the cable when $x = 3$.

Interpret this result.

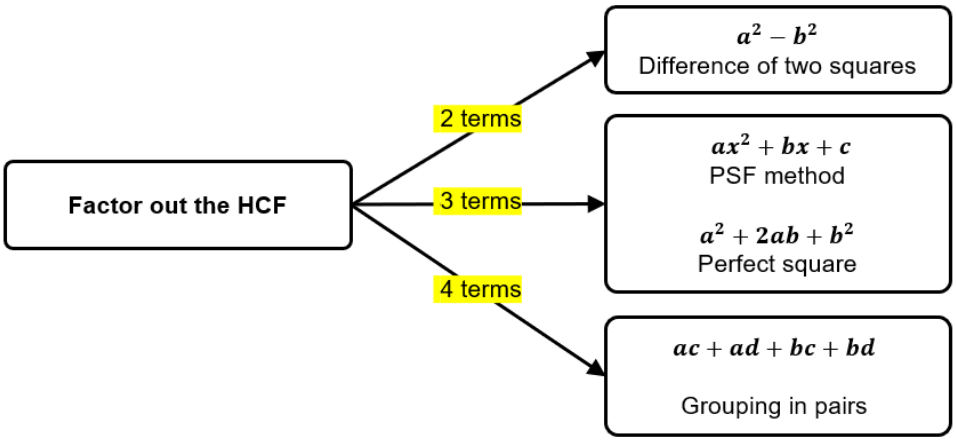
-6, the cable is 6 m below the water.

c Determine where the cable is at the level of the water surface.

$x = 2$ and $x = 5$

So 2 m and 5 m from the side of the channel

Mixed Factorisation Practice



FOUNDATION

1 Factorise the following expressions.

a	$a^2 - 25$	**b**	$b^2 - 25b$	**c**	$c^2 - 25c + 100$
	$(a + 5)(a - 5)$		$b(b - 25)$		$(c - 5)(c - 20)$
d	$2d^2 + 25d + 50$	**e**	$e^3 + 5e^2 + 5e + 25$	**f**	$16 - f^2$
	$(d + 10)(2d + 5)$		$(e + 5)(e^2 + 5)$		$(4 + f)(4 - f)$
g	$16g^2 - g^3$	**h**	$h^2 + 16h + 64$	**i**	$i^2 - 16i - 36$
	$g^2(16 - g)$		$(h + 8)^2$		$(i - 18)(i + 2)$
j	$5j^2 + 16j - 16$	**k**	$4k^2 - 16k - 9$	**l**	$2k^3 - 16k^2 - 3k + 24$
	$(5j - 4)(j + 4)$		$(2k + 1)(2k - 9)$		$(k - 8)(2k^2 - 3)$
m	$2a^2 + ab - 4a - 2b$	**n**	$6m^3n^4 + 9m^2n^5$	**o**	$49p^2 - 121q^2$
	$(a - 2)(2a + b)$		$3m^2n^4(2m + 3n)$		$(7p + 11q)(7p - 11q)$
p	$r^2 - 12r + 32$	**q**	$3t^2 + 2t - 40$	**r**	$5t^2 + 54t + 40$
	$(r - 4)(r - 8)$		$(3t - 10)(t + 4)$		$(5t + 4)(t + 10)$
s	$5t^2 + 33t + 40$	**t**	$5t^3 + 10t^2 + 15t$	**u**	$u^2 + 15u - 54$
	$(5t + 8)(t + 5)$		$5t(t^2 + 2t + 3)$		$(u + 18)(u - 3)$
v	$3x^3 - 2x^2y - 15x + 10y$	**w**	$(p + q)^2 - r^2$	**x**	$4a^2 - 12a + 9$
	$(x^2 - 5)(3x - 2y)$		$(p + q + r)(p + q - r)$		$(2a - 3)^2$

2 Factorise:

a $x^2 - 25$

$(x + 5)(x - 5)$

b $x^2 - 64$

$(x + 8)(x - 8)$

c $4x^2 - 1$

$(2x + 1)(2x - 1)$

d $9x^2 - 16$

$(3x + 4)(3x - 4)$

e $25 - x^2$

$(5 + x)(5 - x)$

f $x^2 + 8x + 16$

$(x + 4)^2$

g $x^2 - 8x + 16$

$(x - 4)^2$

h $9x^2 + 6x + 1$

$(3x + 1)^2$

i $4x^2 - 20x + 25$

$(2x - 5)^2$

j $x^2 + 9x + 14$

$(x + 7)(x + 2)$

k $x^2 - 9x + 14$

$(x - 7)(x - 2)$

l $x^2 + 5x - 14$

$(x + 7)(x - 2)$

m $x^2 - 5x - 14$

$(x - 7)(x + 2)$

n $2x^2 + 5x + 3$

$(2x + 3)(x + 1)$

o $3x^2 - 7x + 2$

$(3x - 1)(x - 2)$

p $6x^2 + x - 2$

$(3x + 2)(2x - 1)$

q $x^2 + 3x + 2x + 6$

$(x + 2)(x + 3)$

r $2x^2 + 6x - x - 3$

$(2x - 1)(x + 3)$

s $ax + bx - ay - by$

$(x - y)(a + b)$

t $2x^2 - 18$

$2(x + 3)(x - 3)$

u $3x^2 + 6x + 3$

$3(x + 1)^2$

v $4x^2 - 4x - 24$

$4(x - 3)(x + 2)$

w $5x^3 - 5x$

$5x(x + 1)(x - 1)$

x $-2x^2 + 8x - 6$

$-2(x - 1)(x - 3)$

3 Factorise the following expressions. Take out a common factor first.

a $3a^2 - 12$

$3(a+2)(a-2)$

b $x^3 - x$

$x(x+1)(x-1)$

c $25y - y^3$

$-y(y+5)(y-5)$

d $9d^2 - 2\frac{1}{4}$

$9\left(d+\frac{1}{2}\right)\left(d-\frac{1}{2}\right)$

e $3x^4 - 27p^2$

$3(x^2 + 3p)(x^2 - 3p)$

f $4x^2 + 14x - 30$

$2(2x - 3)(x + 5)$

g $a^4 + a^3 + a^2 + a$

$a(a + 1)(a^2 + 1)$

h $x^3 - 8x^2 + 7x$

$x(x - 1)(x - 7)$

i $ax^2 - a - 2x^2 + 2$

$(a-2)(x+1)(x-1)$

j $5x^2 + 10x + 5$

$5(x + 1)^2$

k $2x^3 - 8x$

$2x(x + 2)(x - 2)$

l $10x^2 - 40x + 40$

$10(x - 2)^2$

4 Finish factorising the following:

a $(x+3)(x^2-16)$

$(x+3)(x+4)(x-4)$

b $(x+3)(x^2+10x+16)$

$(x+3)(x+2)(x+8)$

c $(x+3)(x^2+8x+16)$

$(x+3)(x+4)^2$

d $(x+3)(4x^2+8x+16)$

$4(x+3)(x^2+2x+4)$

e $(x+3)(4x^2+8x+4)$

$4(x+3)(x+1)^2$

f $(x+3)(4x^2-8x+4)$

$4(x+3)(x-1)^2$

g $5(x+3)(4x^2+8x-5)$

$5(x+3)(2x-1)(2x+5)$

h $5(x+3)(4x^2+8x-12)$

$20(x+3)^2(x-1)$

i $5(x+3)(4x^2+8x)$

$20x(x+3)(x+2)$

5 Factorise.

a $4a^3 - 36a$

$4a(a+3)(a-3)$

b $2x^2 - 18$

$2(x+3)(x-3)$

c $3p^2 - 3p - 36$

$3(p+3)(p-4)$

d $5y^2 - 5$

$5(y+1)(y-1)$

e $5a^2 - 10a + 5$

$5(a-1)^2$

f $3z^3 + 27z^2 + 60z$

$3z(z+4)(z+5)$

g $9ab - 4a^3b^3$

$ab(3+2ab)(3-2ab)$

h $x^3 - x$

$x(x+1)(x-1)$

i $6x^2 + 8x - 8$

$2(x+2)(3x-2)$

j $y^2(y+5) - 16(y+5)$

$(y+5)(y+4)(y-4)$

k $x^4 + 8x^3 - x^2 - 8x$

$x(x+8)(x+1)(x-1)$

l $y^6 - 4$

$(y^3-2)(y^2+2)$

m $x^3 - 3x^2 - 10x$

$x(x+2)(x-5)$

n $x^3 - 3x^2 - 9x + 27$

$(x-3)(x+3)(x-3)$

o $4x^2y^3 - y$

$y(2xy+1)(2xy-1)$

p $24 - 6b^2$

$6(2+b)(2-b)$

q $18x^2 + 33x - 30$

$3(2x+5)(3x+2)$

r $3x^2 - 6x + 3$

$3(x-1)^2$

s $49y^2 + 14y + 1$

$(7y+1)^2$

t $9y^2 - 30y + 25$

$(3y-5)^2$

u $16k^2 - 24k + 9$

$(4k-3)^2$

v $9 - 16x^2$

$(3+4x)(3-4x)$

w $36x^2 - y^2$

$(6x+y)(6x-y)$

x $4a^2 - 81b^2$

$(2a+9b)(2a-9b)$

6 Factorise.

a $t^2 + t + \frac{1}{4}$

$\left(t + \frac{1}{2}\right)^2$

b $x^2 - \frac{4x}{3} + \frac{4}{9}$

$\left(x - \frac{2}{3}\right)^2$

c $9y^2 + \frac{6y}{5} + \frac{1}{25}$

$\left(3y + \frac{1}{5}\right)^2$

d $x^2 + 2 + \frac{1}{x^2}$

$\left(x + \frac{1}{x}\right)^2$

e $25k^2 - 20 + \frac{4}{k^2}$

$\left(5k - \frac{2}{k}\right)^2$

f $(x+2)^2 - y^2$

$(x + 2 + y)(x + 2 - y)$

g $(a-1)^2 - (b-2)^2$

$(a + b - 3)(a - b + 1)$

h $z^2 - (1+w)^2$

$(z + w + 1)(z - w - 1)$

i $(x+2)^2 - (2y+1)^2$

$(x + 2y + 3)(x - 2y + 1)$

j $x^4 - 1$

$(x^2 + 1)(x + 1)(x - 1)$

k $9x^6 - 4y^2$

$(3x^3 + 2y)(3x^3 - 2y)$

l $x^4 - 16y^4$

$(x^2 + 4y^2)(x + 2y)(x - 2y)$

m $125a^2b^2 - 20v^2$

$5(5ab + 2v)(5ab - 2v)$

n $16 - a^4$

$(4 + a^2)(2 + a)(2 - a)$

o $54g^3p - 6gp^3$

$6gp(3g + p)(3g - p)$

p $28c - 63cw^2$

$7c(2 + 3w)(2 - 3w)$

q $144e^5 - 36e^3$

$36e^3(2e + 1)(2e - 1)$

r $x^4 - y^4$

$(x^2 + y^2)(x + y)(x - y)$

s $100k^2 - 6\frac{1}{4}$

$\left(10k + \frac{5}{2}\right)\left(10k - \frac{5}{2}\right)$

t $108m^3n^2 - 27m$

$27m(2mn + 1)(2mn - 1)$

u $c^2 - 1\frac{7}{9}$

$\left(c + \frac{4}{3}\right)\left(c - \frac{4}{3}\right)$

7 Factorise.

a $\frac{1}{25} - x^2$

$\left(\frac{1}{5} + x\right)\left(\frac{1}{5} - x\right)$

b $\frac{64b^2}{25} - \frac{4y^2}{81}$

$\left(\frac{8b}{5} + \frac{2y}{9}\right)\left(\frac{8b}{5} - \frac{2y}{9}\right)$

c $3p^2 - \frac{48}{p^2}$

$3\left(p + \frac{4}{p}\right)\left(p - \frac{4}{p}\right)$

d $(m+3)^2 - 25$

$(m+8)(m-2)$

e $(x-y)^2 - y^2$

$x(x-2y)$

f $(p+x)^2 - (p-x)^2$

$4px$

g $\frac{4c^2}{9} - \frac{e^2}{36}$

$\left(\frac{2c}{3} + \frac{e}{6}\right)\left(\frac{2c}{3} - \frac{e}{6}\right)$

h $4g^2 - (g-2k)^2$

$(3g-2k)(g+2k)$

i $5w^5 - 80w$

$5w(w-2)(w+2)(w^2+4)$

j $a^2(a+2) - 4(a+2)$

$(a+2)(a+2)(a-2)$

k $6x^2 - 5x - 6$

$(3x+2)(2x-3)$

l $6x^2 - 5xy - 6y^2$

$(3x+2y)(2x-3y)$

m $x(y-1) + 3(y-1)$

$(y-1)(x+3)$

n $3x(y-2) + 5(2-y)$

$(y-2)(3x-5)$

o $(m-n)^2 - (m-n)$

$(m-n)(m-n-1)$

a $x^3 - 4x$

$x(x+2)(x-2)$

b $2x^3 + x^2 - 8x - 4$

$(x+2)(x-2)(2x+1)$

c $x^3 - x^2 - 4x + 4$

$(x+2)(x-2)(x-1)$

d $3x^3 + 6x^2 - 3x - 6$

$3(x+1)(x-1)(x+2)$

e $(x+2)^2 - 9$

$(x+5)(x-1)$

f $(x-5)^2 - 4$

$(x-3)(x-7)$

Challenge Exercise

1 Consider this pattern when squaring numbers ending in .5

$1.5^2 = 2.25$ $\quad$ $2.5^2 = 6.25$ $\quad$ $7.5^2 = 56.25$

$1 \times 2 + 0.25 = 2.25$ $\quad$ $2 \times 3 + 0.25 = 6.25$ $\quad$ $7 \times 8 + 0.25 = 56.25$

a How do the whole numbers in the multiplication relate to the original decimal being squared?

The two numbers being used in the multiplication are the number before the decimal and the next consecutive number.

b Use the pattern to find the following. Check your answers using a calculator.

$3.5^2 = 3 \times \square + 0.25$ $= 12.25$ 4	$4.5^2 = \square \times \square + 0.25$ $= 20.25$ 4, 5
$5.5^2 = \square \times \square + 0.25$ $=$ 5, 6, 30.25	$6.5^2 = \square \times \square + 0.25$ $=$ 6, 7, 42.25
$9.5^2 = \square \times \square + \square$ $=$ 8, 19, 0.25, 90.25	$10.5^2 =$ 110.25

c Prove algebraically that this pattern exists for all decimals ending in .5.

$(x + 0.5)^2 =$

$(x + 0.5)^2 = x^2 + x + 0.25 = x(x + 1) + 0.25$

d Prove algebraically that the pattern $25^2 = 20 \times 30 + 25$, $205^2 = 200 \times 210 + 25$ … etc. exists for all numbers ending in 5.

$(x + 5)^2 = x^2 + 10x + 25 = x(x + 10) + 25$

2 Consider that that for any two consecutive integers, the difference of their squares is their sum.

$4^2 - 3^2 = 4 + 3, \quad 5^2 - 4^2 = 5 + 4, \quad 10^2 - 9^2 = 10 + 9,$

Prove that this is always true using algebra.

$(n + 1)^2 - n^2$
$= n^2 + 2n + 1 - n^2$
$= 2n + 1$
$= (n + 1) + n$

3 Consider that for any three consecutive integers, the product of the smallest integer and the largest integer, minus the square of the middle integer, is always −1.

$3 \times 5 - 4^2 = -1, \quad 11 \times 13 - 12^2 = -1, \quad 18 \times 20 - 19^2 = -1$

Prove that this is always true using algebra.

$(n-1)(n+1) - n^2$
$= n^2 - 1 - n^2$
$= -1$

4 "The square of a number is the product of one more and one less than the number, add one."
Show this is true for 10^2

$10^2 = 11 \times 9 + 1$

Prove this is always true using algebra.

WTP $n^2 = (n+1)(n-1) + 1$

RHS $= n^2 - 1 + 1$
$= n^2$
$=$ LHS

5 Factorise $(1 + \beta + \alpha)^2 + 2\alpha\beta(1 + \beta + \alpha) + \alpha^2\beta^2$

This is a perfect square.
$\left((1 + \alpha + \beta) + \alpha\beta\right)^2$

6 Factorise fully:

a $x^4 - 1$

$(x^2 + 1)(x + 1)(x - 1)$

b $x^4 - 16$

$(x^2 + 4)(x + 2)(x - 2)$

c $x^4 - 81$

$(x^2 + 9)(x + 3)(x - 3)$

d $16x^4 - 1$

$(4x^2 + 1)(2x + 1)(2x - 1)$

e $x^4 - y^4$

$(x^2 + y^2)(x + y)(x - y)$

7 Read the example for factorising $x^2 + 5y - y^2 + 5x$

Worked Solution	Explanation
$x^2 + 5y - y^2 + 5x$	How did rearranging help us recognise a difference of two squares?
$= x^2 - y^2 + 5y + 5x$	You see the pattern $a^2 - b^2$
$= (x + y)(x - y) + 5(x + y)$ ←	What happened to get to this line?
$= (x + y)(x - y + 5)$	Factorise by grouping in pairs

Use a similar method to factorise these.

a $x^2 + 7x + 7y - y^2$

$(x - y + 7)(x + y)$

b $x^2 - 2x - 2y - y^2$

$(x - y - 2)(x + y)$

c $4x^2 + 4x + 6y - 9y^2$

$(2x - 3y + 2)(2x + 3y)$

d $25y^2 + 15y - 4x^2 + 6x$

$(5y + 2x)(5y - 2x + 3)$

8 Factorise.

a $(x - 3)^2 - (x + 1)^2$

$-8(x - 1)$

b $16(x + 1)^2 - 9(x - 2)^2$

$(7x - 2)(x + 10)$

c $x^2 - y^2 + x - y$

$(x - y)(x + y + 1)$

d $x^2 - 4y^2 + 6x + 9$

$(x + 3 + 2y)(x + 3 - 2y)$

e $6x^2 + 7x - 3$

$(3x - 1)(2x + 3)$

f $12x^2 - x - 6$

$(4x - 3)(3x + 2)$

9 Factorise fully:

a $x^4 - 5x^2 + 4$

$(x + 1)(x - 1)(x + 2)(x - 2)$

b $x^4 - 13x^2 + 36$

$(x + 2)(x - 2)(x + 3)(x - 3)$

c $9x^4 - 37x^2 + 4$

$(3x + 1)(3x - 1)(x + 2)(x - 2)$

d $4x^4 - 13x^2y^2 + 9y^4$

$(2x + 3y)(2x - 3y)(x + y)(x - y)$

10 Factorise fully:

a $9(x - 1)^2 - 4(x + 2)^2$

$(5x + 1)(x - 7)$

b $25(a + b)^2 - 4(a - b)^2$

$(7a + 3b)(3a + 7b)$

c $x^8 - 1$

$(x^4 + 1)(x^2 + 1)(x + 1)(x - 1)$

d $x^4 - y^4 + x^2 - y^2$

$(x + y)(x - y)(x^2 + y^2 + 1)$

e $a^2(a + 1) - b^2(a + 1)$

$(a + 1)(a + b)(a - b)$

f $(x^2 + x)^2 - (x^2 + x) - 6$

$(x^2 + x - 3)(x^2 + x + 2)$

11

a Show that $n(n + 2) + 1 = (n + 1)^2$ for all integers *n*.

$n(n + 2) + 1 = n^2 + 2n + 1 = (n + 1)^2$

b Hence state the relationship between the product of two integers differing by 2 and the square of the integer between them.

The product of two integers differing by 2 is always 1 less than the square of the integer between them.

c Verify for 11 × 13.

11 × 13 = 143 = 144 − 1 = 12² − 1

12

a Show that $(n + 1)^2 - (n - 1)^2 = 4n$ for all integers *n*.

$(n + 1)^2 - (n - 1)^2 = (n + 1 + n - 1)(n + 1 - n + 1) = (2n)(2) = 4n$

b Hence show that the difference of the squares of any two integers differing by 2 is always a multiple of 4.

Since $4n$ is a multiple of 4 for all integers *n*, the difference of squares of any two integers differing by 2 is always divisible by 4.

13

a Show that $(2n + 1)^2 - (2n - 1)^2 = 8n$ for all integers *n*.

$(2n + 1)^2 - (2n - 1)^2 = [(2n + 1) + (2n - 1)][(2n + 1) - (2n - 1)] = (4n)(2) = 8n$

b Hence show that the difference of the squares of any two consecutive odd integers is always divisible by 8.

Since $8n$ is divisible by 8 for all integers *n*, the difference of squares of any two consecutive odd integers is always divisible by 8.

14

a Show that $(2n + 1)^2 = 8k + 1$ for some integer *k* that depends on n. Hence explain why the square of any odd integer always leaves remainder 1 when divided by 8.

$(2n + 1)^2 = 4n^2 + 4n + 1 = 4n(n + 1) + 1$

One of n or $(n + 1)$ is always even (consecutive integers), so $n(n + 1) = 2m$ for some integer m.

Therefore $4n(n + 1) = 8m$, giving $(2n + 1)^2 = 8m + 1$.

The square of any odd integer leaves remainder 1 when divided by 8.

b Verify for 7^2 and 11^2.

$7^2 = 49 = 48 + 1 = 8 \times 6 + 1$

$11^2 = 121 = 120 + 1 = 8 \times 15 + 1$

15 Show that:

a $n^3 - n$ is divisible by 6 for all integers n.
(hint: a number is divisible by 6 if it is divisible by 2 and 3)

$n^3 - n = n(n+1)(n-1)$

$n^3 - n$ is the product of 3 consecutive integers; one of the integers must be even, and one must be divisible by 3. Therefore, divisible by 6.

b $n^4 + 2n^3 - n^2 - 2n$ is divisible by 24 for all integers n.

$n^4 + 2n^3 - n^2 - 2n = n(n+1)(n-1)(n+2)$; a product of 4 consecutive integers.

Two of the numbers are even, and one of them must be divisible by 4.

One of the numbers must be divisible by 3.

Therefore, divisible by 24 as it is divisible by 8 (from the two even numbers, one divisible by 4) and 3.

Note: it is not enough to prove divisibility by 4 and 6, as they are not coprime (e.g. 12 is divisible by 4 and 6 but not 24)

8 Prove that every square number divided by 4 leaves a remainder of 1 or 0.

Every integer is either even $(2n)$ or odd $(2n+1)$.

$(2n)^2 = 4n^2$, remainder 0.

$(2n+1)^2 = 4n^2 + 4n + 1$, remainder 1.

9 N is a number divisible by 5. Prove that $(N+1)^2 + (N+2)^2$ is also divisible by 5.

Write $N = 5L$. $(N+1)^2 + (N+2)^2$

$= 2N^2 + 6N + 5$

$= 2(5L)^2 + 6(5L) + 5$

$= 5(2 \times 5L^2 + 6L + 1)$.

Therefore $(N+1)^2 + (N+2)^2$ is divisible by 5.

10 Prove the following algebraically.

a Any even number added to another even number is an even number.

Write even numbers as $2k$ and $2m$.

$2k + 2m = 2(k + m)$, which is even.

b The sum of two odd numbers is an even number.

Write odd numbers as $2k + 1$ and $2m + 1$.

$(2k + 1) + (2m + 1) = 2(k + m + 1)$, which is even.

c The product of two odd numbers is an odd number.

$(2k + 1)(2m + 1)$

$= 4km + 2k + 2m + 1$

$= 2(2km + k + m) + 1$, which is odd.

d The sum of an integer and its square is even. Hint: consider integers either as odd or even.

Case 1: integer is odd, $2k + 1$.

$(2k + 1) + (2k + 1)^2$

$= 4k^2 + 6k + 2$

$= 2(2k^2 + 3k + 1)$, which is even.

Case 2: integer is even, $2k$.

$2k + (2k)^2$

$= 2k + 4k^2$

$= 2(k + 2k^2)$, which is even.

11 If $p^2 = p + 1$, find p^3 in terms of p.

$p^3 = p^2 \times p$

$= (p + 1)p$

$= p^2 + p$

$= (p + 1) + p$

$= 2p + 1.$

12 If $x + y = 1$ and $x^2 + y^2 = 5$, find the value of:

a $(x + y)^2$

$(x + y)^2 = 1^2 = 1.$

b xy

$(x + y)^2 = x^2 + y^2 + 2xy,$

$\therefore 1 = 5 + 2xy,$

$\therefore xy = -2.$

13 Look for a pattern with each of the following scenarios by trying a few numbers, come up with a hypothesis and prove your findings algebraically.

a Difference between the squares of two odd numbers.

Let odd numbers be $2n+1$ and $2m+1$. $(2n+1)^2-(2m+1)^2=4(n^2+n-m^2-m)$.

The difference is always a multiple of 4.

Alternatively, $(2n+1)^2-(2m+1)^2=(\text{odd}-\text{odd})(\text{odd}+\text{odd})=\text{even}\times\text{even}$, a multiple of 4.

b Difference between the squares of consecutive even numbers.

Let consecutive even numbers be $2m$ and $2m+2$. $(2m+2)^2-(2m)^2=4(2m+1)$.

The difference is always a multiple of 4.

c Difference between consecutive square numbers.

Let consecutive square numbers be m^2 and $(m+1)^2$. $(m+1)^2-m^2=2m+1$.

The difference is always an odd number.

14 If $k^2=k+5$, write an expression in terms of k and a constant only for:

a k^3

$k^3=k\times k^2$

$=k(k+5)$

$=k^2+5k$

$=(k+5)+5k$

$=6k+5.$

b k^4

Method 1: $k^3\times k$

$=(6k+5)k$

$=6k^2+5k$

$=6(k+5)+5k$

$=11k+30.$

Method 2: $k^2\times k^2$

$=(k+5)^2$

$=k^2+10k+25$

$=(k+5)+10k+25$

$=11k+30.$

15 If $x + y = 9$ and $xy = 5$, find the value of:

a $x^2 + y^2$

$(x + y)^2 = x^2 + 2xy + y^2 = 81,$

$x^2 + y^2 = 81 - 10 = 71.$

b $(x - y)^2$

$(x - y)^2 = x^2 - 2xy + y^2$

$= 71 - 10 = 61.$

16 Using the fact that $x^2 - y^2 = (x - y)(x + y)$, factorise the following:

a $x^4 - 16$

$x^4 - 16$

$= (x^2 - 4)(x^2 + 4)$

$= (x - 2)(x + 2)(x^2 + 4)$

b $(x - 2y)^2 - 9$

$(x - 2y)^2 - 9$

$= [x - 2y - 3][x - 2y + 3]$

c $4 - (x - 5)^2$

$4 - (x - 5)^2$

$= [2 - (x - 5)][2 + (x - 5)]$

$= [7 - x][x - 3]$

d $80 - 5y^4$

$80 - 5y^4$

$= 5(16 - y^4)$

$= 5(2 - y)(2 + y)(4 + y^2)$

e $x^8 - 1$

$x^8 - 1$

$= (x^4 - 1)(x^4 + 1)$

$= (x^2 - 1)(x^2 + 1)(x^4 + 1)$

$= (x - 1)(x + 1)(x^2 + 1)(x^4 + 1)$

f $(x + 3y)^2 - 4y^2$

$(x + 3y)^2 - 4y^2$

$= [x + 3y - 2y][x + 3y + 2y]$

$= [x + y][x + 5y]$

g $x^4 + 4$ (Hint: add and subtract $4x^2$.)

$x^4 + 4 + 4x^2 - 4x^2$

$= (x^2 + 2)^2 - 4x^2$

$= (x^2 + 2 - 2x)(x^2 + 2 + 2x)$

17 Find the four prime factors of $2^{16} - 1$ by factorising.

$2^{16} - 1$

$= (2^8 - 1)(2^8 + 1)$

$= (2^4 - 1)(2^4 + 1)(2^8 + 1)$

$= (2^2 - 1)(2^2 + 1)(2^4 + 1)(2^8 + 1)$

$= 1 \times 3 \times 5 \times 17 \times 257$

Prime factors are 3, 5, 17, 257.

18 Find the last term that makes the following factorable, then factorise.
There are several possibilities for each.

a $x^2 + 4x +$

$x^2 + 4x + 4 = (x + 2)^2$

$x^2 + 4x + 3 = (x + 1)(x + 3)$

b $x^2 - 4x +$

$x^2 - 4x + 4 = (x - 2)^2$

$x^2 - 4x + 3 = (x - 1)(x - 3)$

c $x^2 - 4x -$

$x^2 - 4x - 12 = (x - 6)(x + 2)$

$x^2 - 4x - 32 = (x - 8)(x + 4)$

$x^2 - 4x - 21 = (x - 7)(x + 3)$

Infinite solutions

d $x^2 - x -$

$x^2 - x - 6 = (x - 3)(x + 2)$

$x^2 - x - 12 = (x - 4)(x + 3)$

$x^2 - x - 20 = (x - 5)(x + 4)$

Infinite solutions

e $x^2 - 5x$

$x^2 - 5x - 14 = (x - 7)(x + 2)$

$x^2 - 5x + 6 = (x - 3)(x - 2)$

$x^2 - 5x + 4 = (x - 4)(x - 1)$

19 Without evaluating each square, find the value of:

$$30^2 - 29^2 + 28^2 - 27^2 + 26^2 - 25^2 + \cdots + 4^2 - 3^2 + 2^2 - 1^2$$

Group consecutive pairs using $x^2 - y^2 = (x - y)(x + y)$:

$(30^2 - 29^2) + (28^2 - 27^2) + \cdots + (2^2 - 1^2)$

$= (30 + 29) + (28 + 27) + \cdots + (2 + 1)$

$= 1 + 2 + 3 + \cdots + 29 + 30$

$= \frac{30 \times 31}{2} = 465$